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Prevalence of malnutrition among old age people in Africa: Systematic Review and Meta-Analysis

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Abstract

Objective: To summarize available evidence and estimate the pooled prevalence of malnutrition among old age people in Africa.

Design: Systematic review and meta-analysis

Setting: Any health care or community research, reporting the prevalence or incidence of malnutrition in Africa from January 2000 to October 2021.

Participants: Old age people, aged above 60 years.

Outcome measures: Malnutrition either under or overnutrition.

Results: A total of 1,442 studies were retrieved based on the search strategy, where only 36 studies (n = 15,266 participants) reported from 11 African countries were included for meta-analysis. The reported prevalence of malnutrition ranges from 2.2 to 77.3% across Africa. Overall, the pooled prevalence of malnutrition was 18% (95% CI: 15-22; $I^2 = 98.1$; $p < 0.001$).

Conclusion: The prevalence of malnutrition in African old age people is high and differs by setting, assessment tool, and country of residence. Hence, due attention to geriatric nutrition is mandatory, and the need for a valid, reliable, and simple screening tool should be thought of.

Strengths and limitations of this study

- This provides a comprehensive estimate of the prevalence of malnutrition among old age people in Africa, including more recent studies conducted in the continent.

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- Looked at the prevalence of malnutrition in the elderly using any of the validated screening tools.
- Only HINARI/PubMed and Google Scholars were retrieved.
- Limited to include only English literature.

Key words: Africa; Malnutrition; Old age people; Prevalence; Meta-analysis; Systematic-review

Introduction

Background

The global demographic structure of the population is changing dramatically, notably in Africa, due to several social and economic milestones (1). These resulted in expanding population, improved life expectancy, and increased population size of older ages (1,2). The absolute number of old age people (60 years and above (3)) is expected to grow more rapidly in the coming decades than in any other part of the world (1).

In Africa, the risk of malnutrition among older people is a major challenge to the health care system and needs special and urgent attention. The usual food production practices, marketing style, and living standards have all contributed to an increase in the use of low-cost, packaged products that are high in fat, energy, and salt and yet low in nutritional quality. More importantly, poor infrastructure and limited resources, combined with conflict and poor access to healthcare services, are factors that contribute to the overwhelming levels of malnutrition and food insecurity on the continent (4). The lack of policy direction for old age people, combined with prevalent chronic illnesses, adds a significant burden to the growing burden of malnutrition (5,6). Due to these and other aggravating factors, the government's commitment to ending all forms of malnutrition by 2030 (7) will be difficult to achieve.

While any age group may suffer from malnutrition, it is most common among old age people due to the changes in physiological, psychosocial, and health characteristics of individuals in this age group (8). Furthermore, malnutrition has serious and life-threatening consequences that are known to be the main causes of increased morbidity and mortality among old age people (9). Besides, complications of malnutrition, such as osteoarthritis, osteoporosis, diabetes, cardiovascular disease, and hypertension, inflict a significant social and economic burden on them (10).

Previous studies reported that the prevalence of malnutrition among old age people varies between

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1.1% and 72.2% in different settings (11–15), where some of the variations could be attributed to differences in the measurement tools, study settings, and demographic groups that have been studied (16,17). Moreover, none of the previous reviews targeted the old age people in Africa (18–21), where the root causes of malnutrition are expected to be large. However, having concrete evidence on the burden of malnutrition using a more rigorous and systematic manner than individual pocket studies is the first step toward enhancing the elderly's interventions and policy direction. Context-specific evidence will provide better inform decision-makers in making informed decisions on addressing malnutrition among old age people for a better quality of life.

Objective

To estimate the prevalence of malnutrition among old age people in Africa according to the assessment tools, study setting, and regional location.

Methods

Search strategies

Before the beginning of this systematic review, we conducted a systematic search of review papers from the Cochrane Library, the International Prospective Register of Systematic Reviews (PROSPERO), and the Joanna Briggs Institute (JBI), and no review had been conducted on malnutrition among old age people in Africa. Yet, review protocol was not registered. Relevant publications were searched systematically from HINARI, PubMed, and Google Scholar sources from 01 January 2000 to 16 October 2021. We employed the search through relevant combinations of MeSH terms and related keywords, such as "malnutrition", "obesity", "overweight", "old age people", "older adults", "elderly", and "Africa" as indicated in **Supplementary File**. Lists of references from relevant studies were also manual-searched. Countries or territories included in Africa were defined according to the United Nations (UN) classification (22). Studies were imported and checked for duplicates in Mendeley Desktop (Version 1.19.3).

Eligibility criteria

Inclusion criteria

Based on the condition, context, and population (CoCoPop) framework (23) we included research publications in English language from January 1, 2000 to October 16, 2021 that reported the prevalence of malnutrition among old age persons aged 60 years and up in African countries.

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Intending to get a relatively up-to-date prevalence estimate and obtain a comparable age and malnutrition definition, we restricted studies conducted starting in 2000. All studies that reported the prevalence and/or incidence of malnutrition among old age people were considered regardless of the study design. When multiple studies reported the prevalence of malnutrition based on the same study, the one with the larger sample size was included.

Exclusion criteria

Qualitative studies, study protocols, abstracts, opinions, commentaries, case reports, review papers, and studies on African-Americans who do not live on the African continent were all excluded from the review. Publications reporting malnutrition in post-intervention were excluded. Duplicated articles and articles without access to the full text were removed from the review.

Outcome measuring tools

Studies that measured malnutrition with any standard and validated tools, including questionnaires such as Patient-Generated Subjective Global Assessment (PG-SGA), Mini Nutritional Assessment (MNA), Subjective Global Assessment tools (SGA), or Nutritional Screening Checklist (NCL), anthropometric [mid-upper arm circumference (MUAC), calf-circumference (CC), body mass index (BMI), etc.], or biochemical indices (serum haemoglobin, albumin, etc.) were considered. We used the reported prevalence of malnutrition or undernutrition based on the standard definition set for each nutritional assessment method. When the entire text of an article could not be obtained or if there was missing information, the corresponding author(s) was contacted through email. Requested full-text papers that were not submitted until October 31, 2021, and the study was discarded.

Study selection and quality assessment

A two-step process was used to select the relevant publications. First, the scan of the titles and abstracts was conducted for the inclusion of references that meet the eligibility criteria. The entire text was retrieved when any doubts existed about the relevance of a study. Second, the full texts were then used to make a judgement on whether or not the study should be included in the review. The two authors work individually on all components of the selection process. Disagreements were settled through conversation.

Selected studies were evaluated using a ten-item rating checklist designed for research reporting

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prevalence (24), where better quality articles have a higher score as indicated in **Supplementary Table 1**.

Data extraction

The two authors separately extracted data from the full-text articles using a pre-defined standard format produced in Excel. The extract comprised the following information: general (authors, year of publication, and country), participants (age, gender, and morbidity status), and study details (study design, study setting, sample size, measurement tool used, and outcome measures). Morbidity was coded as one or multi when a single or multiple underlying health conditions were an inclusion criterion.

Statistical analysis

The pre-coded extracted data in excel format was exported to STATA software version 14 (25) for descriptive and meta-analysis. For each study, the unadjusted prevalence of malnutrition and standard errors were calculated, and a logarithmic (log) transformation for proportional data was used to reduce variance and obtain the pooled prevalence as an effect size (ES). Considering the level of methodological heterogeneity and variance within the study, a random effect model of the DerSimonian and Laird (26) method was used to conduct the meta-analysis.

The heterogeneity between studies was assessed using a forest plot, Cochran's Q test, and I-square (I^2) test. The I^2 values of 25, 50, and 75% with a *p-value* less than 0.10 represent low, medium, and high heterogeneity, respectively (27). Publication bias was checked graphically in a funnel plot supplemented by Egger's statistics (28). We also applied a subgroup analysis, a sensitivity analysis, and meta-regression to identify the potential sources of heterogeneity. A statistically significant effect size estimate has a P-value of less than 0.05.

Patient and public involvement

This study did not involve any patients or the general public.

Results

Search results

A total of 1,442 articles were obtained through searches, of which 206 were removed due to duplicates. With the title and abstract screening, 971 articles were removed depending on the focus of the review (older age in the African continent), relevant nutritional assessment method, and

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language (other than English). Furthermore, 237 articles were retrieved during the full-text screening. Finally, 37 and 36 articles were selected for qualitative (systematic review) and quantitative analysis (meta-analysis), respectively. The comprehensive selection process presented using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (29) (**Figure 1).**

Characteristics of included studies

A total of 37 studies reported the prevalence of malnutrition among old age people. While 34 studies (92%) were published in the last decades between 2011 and 2020, the remaining 8% were issued between 2001 and 2010. All the included studies were reported from 11 African countries, where more than half (65%, n = 24) were from four countries: Ethiopia, Egypt, Nigeria, and South Africa.

Seven studies (18.9%) were from Ethiopia (30–36) and six studies (16.2%) each from Egypt (37–42) and Nigeria (43–48). While five studies (13.5%) were from South Africa (49–53), four studies (10.8%) were from the Central African Republic (CAR) and the Republic of Congo (54–57) three studies (8.1%) from Ghana (58–60), and two studies (6.06%) were from Tanzania (61,62).

Except for three studies (two cohorts (31,47) and one case-control (44)), all other identified studies were cross-sectional study designs. Malnutrition prevalence was reported in four studies as a comparison between institutional versus noninstitutional (44,61), high versus low economic classes (50), and with and without dementia (57). However, one study from Tanzania examined the nutritional status after a post-supplemental food program (63) and was excluded from the meta-analysis. Majority of the studies (n = 22) were conducted in community settings, while 11 studies were conducted among institutionalized elders.

Moreover, a total of 15,266 study participants were found, of whom more than half (58%) were females. The selected studies had sample sizes ranging from 40 (64) to 2,219 (37) people. Regarding the nutritional assessment tools used, more than half of the studies reported malnutrition based on either the full-or short-form of the Mini Nutritional Assessment tool (MNA-FF: n = 15, MNA-SF: n = 4). While 18 studies employed various anthropometric dimensions such as BMI, MUAC, and CC separately or in combinations to assess malnutrition. One study used an interviewer- administered Nutritional Screening Checklist (NCL)(38) and another study utilized a combination of MNA and the geriatric nutrition risk index (GNRI)(64). Two-thirds (n = 23) of the

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studies reported single (n = 8) or multiple (n = 15) underlying clinical conditions such as fracture, heart disease, dementia, Parkinson's disease, or cancer (**Supplementary Table 2**).

Quality of the studies

According to the quality assessment checklist, studies were either in moderate (6-8 points) (39,40,64,65,41,42,44,46–48,53,61) or high-quality groups (9 or 10 points) (30,31,45,49–52,54,55,57–59,32,60,62,66,33–38,43). Study bias was mostly exacerbated as a result of convenience sampling and a low response rate among study participants. In almost all studies, the possibility of external validity was low as the target population was not nationally representative of the country. However, the quality of two studies (56,63) was not assessed and hence not processed for quantitative data analysis. The first was (56) due to duplication of results from previous data (57) and the latter (63) was due to the outcome variable being measured after an intervention (**Supplementary Table 3**).

Prevalence of malnutrition

The prevalence of malnutrition varied highly across the studies conducted in different countries, assessment tools used, and healthcare settings. We presented an overview of the findings from various studies using two assessment tools in the section below.

Prevalence of malnutrition using BMI

Seventeen studies from hospitals (n = 5), the community (n = 10), a long-term care center (n = 1), and a daycare center (n = 1), utilized body mass index (BMI) as a nutritional assessment tool. As low as zero (53) and 4.0% (49) of research participants respectively from the elderly daycare center and hospital settings in South Africa were classified as undernourished (BMI $\leq$ 18.5 kg/m^2). Furthermore, the highest prevalence of undernutrition (25.6%) was reported in a Tanzanian hospital (62). From the community settings, undernutrition was reported at 9.9% in Ghana (59) and 26.4% in the East Africa (66).

On the other hand, 77% of the elderly people studied in a South African hospital (49) were overweight or obese. There were none from Tanzania (62) in a comparable situation. While 15.3%

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of study participants in the East Africa (66) and 42.1% in Ghana (59) were affected by overweight or obesity in the community settings.

Prevalence of malnutrition using MNA

Old age people in the community were more victims of malnutrition when using MNA compared to hospitals and long-term care centers. The prevalence ranged from 5.7% in Niger (65) to 56% in Egypt (38) and 18% in Egypt (38) to 58.5% in Niger (65), respectively. While 2.24% (48) to 40.46% (47) and 11.8% (45) to 52.7% (47) of old age people all from the Nigerian hospitals were malnourished and at risk of malnutrition, respectively. Besides, 7.3% in South Africa (50) and 10.8% in Egypt (41) were malnourished using MNA from the long-term care center.

Similarly, 10.9% of community study participants in Egypt (37) and 77.3% in Ethiopia (33) were malnourished, while none from Ethiopia(33) and 41.9% from Egypt (37) were at risk of malnutrition using MNA-SF. Moreover, 4.57% of hospital study participants in Nigeria (46) were malnourished, while 95.43% were either normal or at risk of malnutrition.

Meta-analysis

Adequate data were available to conduct pooled malnutrition prevalence estimates for two nutrition screening tools: BMI (n = 16) and MNA (n = 20). Since the full-and short-form of MNA and NCL tools are similar in the interpretation of the outcome variable, they were treated as similar tools. The random pooled prevalence of malnutrition in Africa was 18% (95% CI: 15-22; p < 0. 001) **(Figure 2)**.

As illustrated in Figure 2, a high level of statistical heterogeneity was observed between studies (heterogeneity of Cochran Q statistics = 1828.98 with *df* = 35 and *p-value* < 0.001; I^2 = 98.1%; *Tau-square* (τ^2 = 0.01).

Moreover, a minimal publication bias has been noted in the published studies. Studies with lower standard error and larger sample size might be included, as indicated in of the funnel and Egger's publication plots (**Figure 3**). Smaller studies with larger standard errors plotted on the right of the funnel plot tend to have both smaller and larger odds ratios. This may indicate the existence of bias due to small-study effects. Moreover, the Egger plot shows that the data near the origin are

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unsystematically elevated, where the confidence interval (CI) does not include zero, indicating asymmetry in the funnel plot and evidence of publication bias.

Furthermore, Egger's regression asymmetry test value was 3.50 (p = 0.008), suggesting publication bias. Here, the positive coefficient indicates that small studies overestimate the effect size. However, no study appeared to have a significant impact on the summary effect size in the sensitivity analysis, strongly suggesting that the effect estimate is consistent across groups and potentially giving a reliable estimate of the outcome.

Subgroup analysis and meta-regression

Subgroup analysis and meta-regression were performed for the study regions, settings, assessment tools, age group, and presence of multimorbidity using the random-effects inverse-variance model with the DerSimonian-Laird estimate. In general, a relatively high prevalence of malnutrition was observed in central Africa (3.76%; 95% CI: 3.22-4.39) as compared to other regions of the continent. Similarly, the community elderly (3.12%; 95% CI: 2.66-3.67), older age groups (3.54%; 95% CI: 2.32–5.41), and those with no reported clinical condition (3.16%; 95CI: 2.62–3.80) had a higher malnutrition burden than their counterparts. The prevalence estimates had a statistically significant difference among all subgroups except for the assessment tools used (p = 0.430) (Table 4).

Table 4: Subgroup analysis for the prevalence of malnutrition among old age people, 2021

Subgroup	No of studies included	Prevalence with 95%CI	Cochran's Q statistics for heterogeneity	P value	I^2
Regions					
Northern Africa	8	2.82[2.72-2.92]	3.93	0.788	0.0%
Western Africa	10	2.46[1.95-3.11]	263.06	< 0.001	96.6%
Central Africa	3	3.76[3.22-4.39]	2.08	0.354	3.7%
Eastern Africa	10	3.46[2.29-5.22]	312.49	< 0.001	97.1%
Southern Africa	5	1.79[1.41-2.26]	60.54	< 0.001	93.4%
Between groups			42.08	< 0.001	
Study settings					
Community	21	3.12[2.66-3.67]	829.56	< 0.001	97.6%
Hospital	11	2.12[1.67-2.70]	132.88	< 0.001	92.5%
Long-Term Care	3	2.34[2.08-2.63]	0.31	0.859	0.0%
Elderly day care center	1	1.04[0.87-1.24]	0.00	-	-
Between groups			86.79	< 0.001	

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Assessment Tools					
BMI	16	2.98[2.37-3.75]	213.68	< 0.001	93.0%
NCL	1	5.74[0.71-46.74]	-	-	-
MNA	15	2.45[2.11-2.85]	264.09	< 0.001	94.7%
MNA-SF	4	2.45[1.96-3.06]	136.91	< 0.001	97.8%
Between groups			2.76	0.430	
Age group					
≥ 60	26	2.54[2.17-2.98]	1102.79	< 0.001	97.7%
≥ 65	10	3.54[2.32-5.41]	361.43	< 0.001	97.5%
Between groups			2.06	< 0.001	
Morbidity					
One	7	1.66[1.46-1.89]	91.58	< 0.001	93.4%
Multi	16	2.84[2.50-3.22]	205.87	< 0.001	92.7%
Not reported	13	3.16[2.62-3.80]	36.06	< 0.001	66.7%
Between groups			16.44	< 0.001	
Overall	**37**	**2.77(2.43-3.16)**	**1573.48**	**< 0.001**	**97.7%**

However, only the study setting was statistically significant (adjusted R-squared = 39.2%, p = 0.049) predicting the prevalence of malnutrition in the meta-regression analysis.

Discussion

We conducted this review paper with the intention of estimating the pooled prevalence of malnutrition disaggregated by relevant factors such as study setting, screening tool, and so on. According to the findings, the prevalence of malnutrition ranges from 4.0% (49) to 77.3% (33) across countries, depending on the assessment tools used and the study settings. These prevalence figures are consistent with previous European estimates of 3.8 to 67.4% (19,21) among older people despite differences in the settings. In contrast, the highest risk of malnutrition was observed in the community-dwelling old age people assessed by the MNA (both full and short forms) than in other studies; 7.5% to 77.3% versus 5% (18) to 19% (67). The variation may be due to the economic variables of the countries.

Overall undernutrition was 17.1% in the African continent using BMI, which is comparable to European countries, 3.8-18.2% (21). While 27% of the study participants in the continent were overweight or obese. Furthermore, many studies (n=14) in our review used BMI as a nutritional assessment tool for old age people. Similar studies in Africa still preferred to use BMI for old age people where 19.9% and 30% of subjects were undernourished, overweight or obese, respectively (68).

However, the BMI cutoff point used to identify underweight can have a significant impact on

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prevalence estimates, which tend to misclassify the elderly with physical spinal deformities and other factors that can affect height measurements (69). Due to such measurement errors encountered, the assessment of nutritional status in the old age people become a challenging task, which warrants more comprehensive and multidimensional assessments including clinical examination, anthropometric measures, laboratory tests, dietary surveys, and social aspects (70,71). This means one assessment tool may not be suitable for all settings, which need contextualization (18). Though more than 22 best-validated nutrition screening tools were identified for older adults (17,19), there has long been a lack of consensus regarding the criteria needed to make a diagnosis of malnutrition.

On the other hand, the meta-analysis revealed a pooled prevalence of malnutrition of 18% (95% CI: 15-22; $p < 0.001$). This is in the range of the ones reported from the world, ranging from 0.8 to 24.6% (72) depending on the study regions, settings, and assessment tools used. The increasing number of an ageing population in Africa and nutritional transitions might have contributed to a higher burden of malnutrition. Our estimate is also consistent with the pooled prevalence of Indian studies (18.29%; 95% CI: 15.24-21.57) (73). However, the result is slightly higher than the prevalence of malnutrition reported from the Central Demographic Republic of Congo and Nigeria (14.5%; 95% CI: 0.0-40.4%) (72). This disagreement could be explained by the use of various assessment procedures and the merging of pooled estimates from two countries where the burden of malnutrition might be higher.

Surprisingly, the current prevalence estimate is lower than the reviews reported from European older persons (48.4%; 95% CI: 41.5-51.8) (19). The disparity could be attributed to economic factors, health literacy, and healthcare access, and the majority of the previous review data came from hospital settings, implying that individuals with higher health risks were included.

Still, there was significantly high heterogeneity between the studies ($Q = 1873.99$; $df = 36$; $p < 0.001$; $T^2 = 0.01$; $I^2 = 98.1\%$). Sensitivity analysis did not improve this heterogeneity, which is consistent with other previous findings (18,72). Because of the highly complex physiological, social, and temporal nature of malnutrition, which differs from person to person, as well as bias introduced by research design, analysis of the pooled prevalence of malnutrition is difficult to account for all sources of variation (74).

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In case of subgroup analysis, the frequency of malnutrition showed a great variation depending on the study regions, study settings, age groups, and clinical conditions of study participants. Malnutrition was found to be prevalent in central Africa (3.76%), in older age groups (3.54%), with no documented clinical condition (3.16%), and in community-dwellers (3.12%). The prevalence was statistically significant in all subgroups except between the assessment tools used ($p = 0.430$). These findings are consistent with the reviews from India (73) and the rest of the world (72). While the meta-regression identified the study setting as a statistically significant (adjusted $R^2 = 39.20\%$, $p = 0.049$) predictor of malnutrition prevalence. This is, however, different from the reviews conducted in India (73) and the world (72) where the study region ($R^2= 27.1\%$, $p = 0.026$) and assessment tools ($R^2= 76.47\%$, $p<0.001$) were statistically significant predictors of the malnutrition prevalence, respectively. The variation might be due to the difference in assessment tools used between the studies.

Strength and Limitation

To our knowledge, this is the first systemic review and meta-analysis that provides a comprehensive estimate of the prevalence of malnutrition among old age people in Africa, including more recent studies conducted in the continent. Even though this review has its strengths, it is not without limitations. Due to the inherent heterogeneity among the included studies ($I^2 = 98.08\%$), the interpretation of the current study needs precaution. Being limited to including only English literature and not including research published in books, magazines, and others might limit the generalizability of the estimate.

Conclusion

This systematic review and meta-analysis showed high malnutrition prevalence among old age people which warrants a targeted policy intervention. The prevalence varied greatly depending on the study setting, the tool used, and the country of residence. To better monitor and evaluate, the use of a more reliable, valid, and standardized malnutrition assessment tool is strongly recommended.

List of abbreviations

BMI: Body Mass Index; CC: Calf Circumference; HINARI: Health InterNetwork Access to Research Initiative; MeSH: Medical Subject Headings; MNA: Mini Nutritional Assessment

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Declarations

Ethical approval and formal consent

Not applicable

Consent for publication

Not applicable

Availability of data and materials

All data generated or analyzed during this study are included in this published article and its supplementary information files, tables, and figures.

Competing interests

The authors declare that they have no competing interests

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Author Contributions

The study was created and designed by AM under the supervision and approval of NF. They are both responsible for all parts of this work. The final manuscript was read and approved by both writers.

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Figure Legends

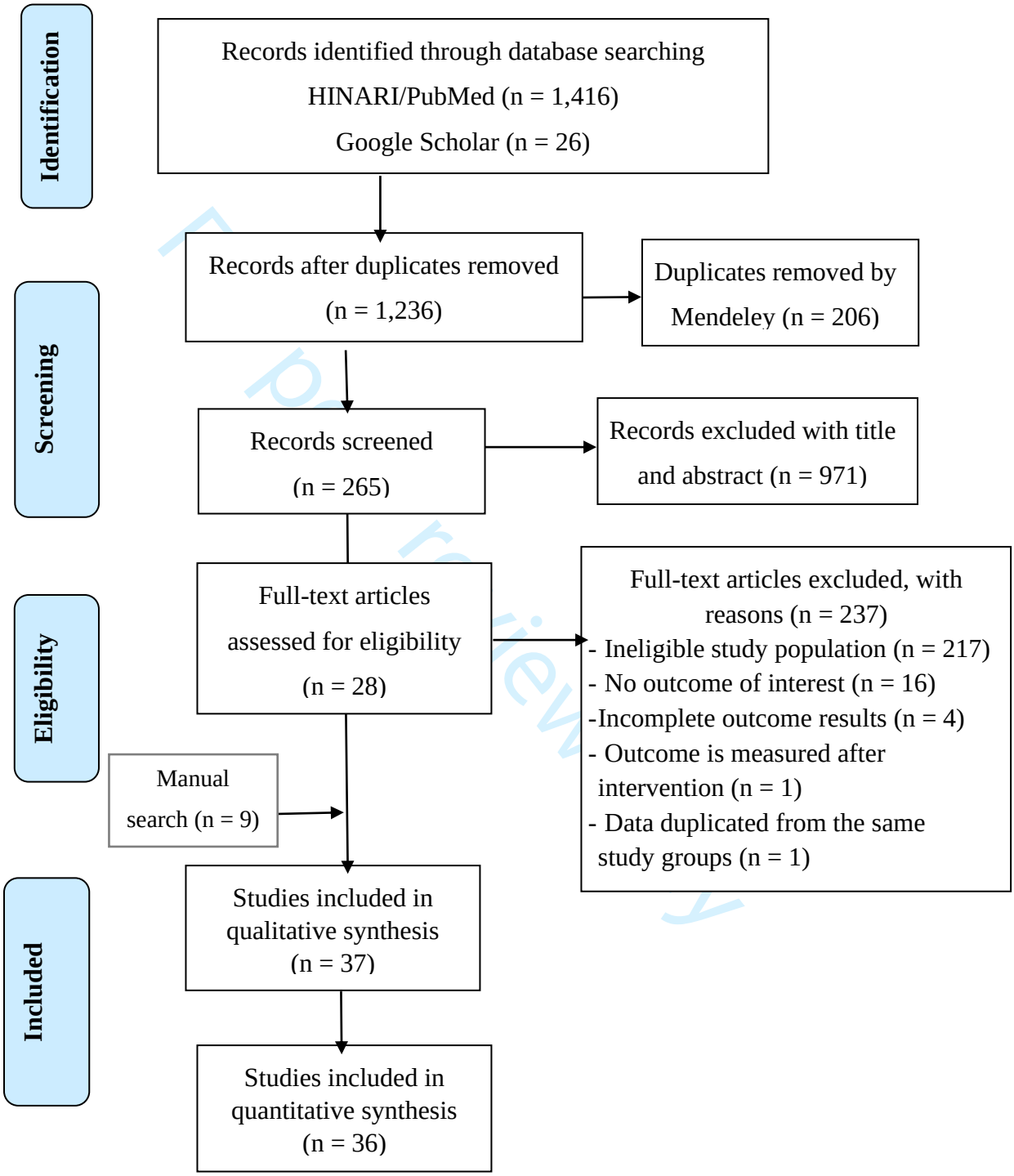


Figure 1: PRISMA flow-chart of studies screened and reviewed.

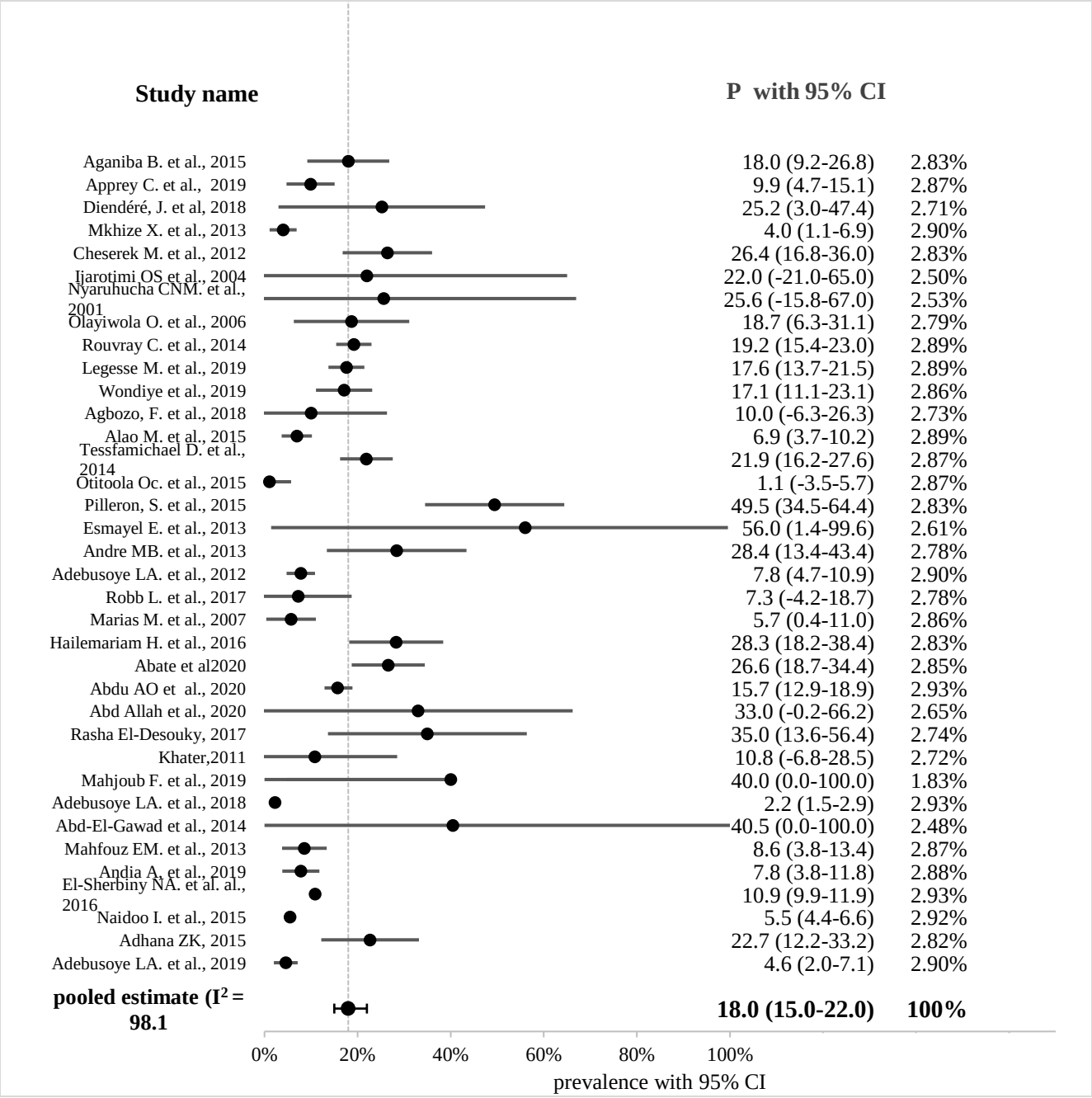


Figure 2: Forest plot displaying the pooled prevalence of malnutrition among old age people in Africa.

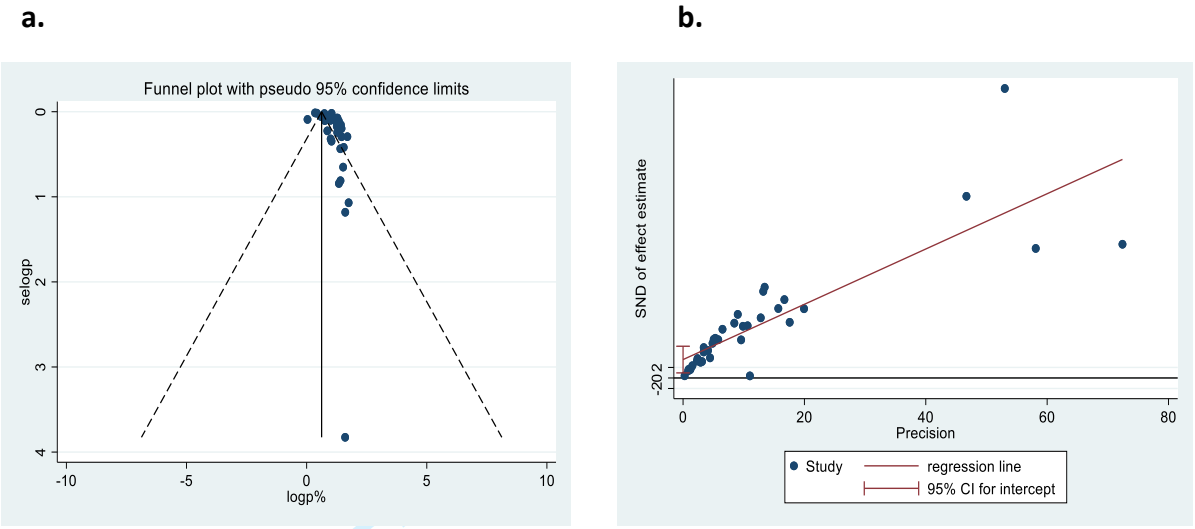


Figure 3: Probability of publication bias for included studies (a) funnel plot and (b) Egger's publication bias plot.

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Supplementary material

Table 1: Quality appraisal checklist for prevalence studies

Items		Quality score
	External validity Criteria	
1.	Was the study's target population a close representation of the national population in relation to relevant variables?	(1 point)
2.	Was the sampling frame a true or close representation of the target population?	(1 point)
3.	Was some form of random selection used to select the sample, OR was census undertaken?	(1 point)
4.	Was the likelihood of nonresponse bias minimal?	(1 point)
		Total (4 points)
	Internal validity Criteria	
5.	Were data collected directly from the subjects (as opposed to a proxy)?	(1 point)
6.	Was an acceptable case definition used in the study?	(1 point)
7.	Was the study instrument that measured the parameter of interest shown to have validity and reliability?	(1 point)
8.	Was the same mode of data collection used for all subjects?	(1 point)
9.	Was the length of the shortest prevalence period for the parameter of interest appropriate?	(1 point)
10.	Were the numerator(s) and denominator(s) for the parameter of interest appropriate?	(1 point)
		Total (6 points)
	Overall total	10 points

Methodological quality

High quality: score > 8

Moderate quality: score 6-8

Low quality: score 0-5

(Source: Hoy et al.(24))

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Table 2. Characteristics of Selected Studies Using Different Nutritional Assessment Tools

S/no	Authors	Country	Study design	Study setting	Sample size	Age group	Gender		Assessment tool used	Nutritional status (outcome variable) in %				Multi-morbidity
							Male	Female		Under-nutrition	At risk of malnutrition	Normal-weight/ Nutrition	Overweight/ obesity	
1.	Aganiba, B. et al., 2015 (62)	Ghana	C/S	Community	400	≥ 65	177	223	BMI	18.0		60.5	21.5	Not reported
2.	Apprey C. et al., 2019 (63)	Ghana	C/S	Community	375	≥ 60	127	248	BMI	9.9		48	30.9/11.2	Multi
3.	Diendéré, J. et al, 2018 (35)	Burkina Faso	Cohort	Hospital	222	≥ 60	121	101	BMI	25.2		55.0	14.9/5.0	One
4.	Mkhize X. et al., 2013 (53)	South Africa	C/S	Hospital	270	≥ 60	46	224	BMI	4.0		19.0	24.0/53.0	One
5.	MJ, C. et al, 2012 (69)	Lake Victoria Basin of East Africa	C/S	Community	537	≥ 60	227	310	BMI	26.4		58.3	10.8/4.5	Not reported
6.	Ijarotimi & Keshinro, 2004 (65)	Tanzania	C/S	Hospital	100	≥ 60	51	49	BMI	22.0		61.0	17.0	Not reported
7.	Nyaruhucha CNM. et al, 2001 (66)	Tanzania	C/S	Hospital	121	≥ 60	67	54	BMI	25.6		74.4	0	Not reported
8.	Olayiwola O.et al, 2006 (47)	Nigeria	C/S	Hospital	295	≥ 60	153	142	BMI	18.7		60.7	15.7/4.9	Not reported
9.	Jésus p. et al., 2017 (60)	CAR & ROC	C/S	Community	990	≥ 65	405	585	BMI	19.2		52.9	19.1/8.8	Multi
10.	Rouvray C. et al., 2014 (61)	CAR & ROC	C/S	Community	990	≥ 65	405	585	BMI	19.2		52.9	19.1/8.8	Multi
11.	Legesse M. et al., 2019 (34)	Ethiopia	C/S	Community	892	≥ 65	365	527	BMI	17.6		82.4		Multi
12.	Wondiye K et al., 2019 (35)	Ethiopia	C/S	Community	554	≥ 60	295	259	BMI	17.1		80.6	2.3	Not reported
13.	Agbozo, F. et al., 2018 (64)	Ghana	C/S	Community	120	60-70	42	78	BMI	10.0		51.7	21.7/16.6	Not reported
14.	Alao M. et al., 2015 (48)	Nigeria	Case-control	LTC	418	≥ 60	191	227	BMI	6.9		53.3	24.6/15.1	Not reported
15.	Tessfamichael D. et al., 2014 (40)	Ethiopia	C/S	Community	757	≥ 65	194	563	BMI	21.9		78.1		Multi
16.	Otitoola Oc. Et al., 2015 (57)	South Africa	Cohort	Elderly day care center	46	≥ 60	8	38	BMI	0		18.4	80.4	Not reported
17.	Pilleron, S. et al., 2015 (58)	CAR & ROC	C/S	Community	645	> 65	Not reported	Not reported	BMI	68.5		31.5		Not reported
18.	Esmayel EM. et al, 2013 (42)	Egypt	C/S	Community	200	≥65	112	88	NCL	56.0	18.0	26.0		Not reported
19.	Andre MB. et al., 2013 (59)	ROC	C/S	Community	370	≥60	194	176	MNA	13.8	28.4	57.8		Multi
20.	Adebusoye L. A. et al., 2012 (49)	Nigeria	C/S	Hospital	500	≥60	176	324	MNA	7.8	11.8	80.4		Multi
21.	Robb L. et al., 2017 (54)	South Africa	C/S	LTC	124	≥60	35	89	MNA	7.3	55.6	37.1		Not reported
22.	El-Sherbiny N.A. et al., al.2016 (41)	Egypt	C/S	Community	2,219	≥60	1165	1054	MNA-SF	10.9	41.9	47.2		Multi

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23.	Naidoo I. et al., 2015 (56)	South Africa	C/S	Community	984	≥60	224	760	MNA SF	51.0	5.5	43.4		Multi
24.	Adhana, Z K et al., 2015 (37)	Ethiopia	C/S	Community	423	≥60	99	324	MNA-SF	77.3	0	22.7		Multi
25.	Adebusoye L. et al., 2019 (50)	Nigeria	C/S	Hospital	350	>60	Not reported	Not reported	MNA-SF	4.57	95.43			Multi
26.	Marias ML. et al., 2007 (55)	South Africa	C/S	Community	210	≥60	53	157	MNA	5.7	47.1	47.1		Not reported
27.	Hailemariam H. et al., 2016 (36)	Ethiopia	C/S	Community	548	≥60	261	287	MNA	9.3	28.3	62.4		Not reported
28.	Abate et al., 2020 (38)	Ethiopia	C/S	Community	662	≥65	240	422	MNA	26.6	73.4			One
29.	Abdu AO et al., 2020 (39)	Ethiopia	C/S	Community	592	≥65	291	301	MNA	15.7	51.7	32.6		One
30.	Abd Allah et al., 2020 (43)	Egypt	C/S	Hospital	194	≥60	107	87	MNA	33.0	51.5	15.5		Multi
31.	Rasha El-Desouky 2017 (44)	Egypt	C/S	Community	320	≥60	167	153	MNA	35.0	38.4	26.6		Multi
32.	Khater MS et al., 2011 (45)	Egypt	C/S	LTC	120	≥60	56	64	MNA	10.8	40.8	48.4		Multi
33.	F. Mahjoub et al., 2019 (68)	Tunis	C/S	Hospital	40	≥65	21	19	MNA	40.0	52.5	7.5		One
34.	Adebusoye, LA. et al., 2018 (52)	Nigeria	C/S	Hospital	624	≥60	246	378	MNA	2.24	32.69	65.06		One
35.	Abd-El-Gawad et al., 2014 (51)	Nigeria	Cohort	Hospital	131	≥60	67	64	MNA	40.46	52.67	6.87		Multi
36.	Mahfouz, E.M. et al., 2013 (46)	Egypt	C/S	Community	350	≥60	136	214	MNA	8.6	29.7	61.7		Multi
37.	Andia A, et al., 2019 (70)	Niger	C/S	Community	384	≥60	132	252	MNA	7.8	58.5	33.7		Multi

Abbreviations: BMI = Body mass index
CAR = Central Africa Republic
ROC= Republic of Congo
C/S: Cross-section
MNA: Mini Nutritional Assessment Tool
MUAC: mid-upper arm circumference
LTC: Long term care
MNA-SF: Mini Nutritional Assessment Tool- short Form
NCL: Nutritional Screening Checklist

Table 3. Summary of the quality of included studies

S/no	Author & year of publication	Country	External validity quality Score				Internal validity quality Score						Total Score	Study Quality
			1	2	3	4	5	6	7	8	9	10		
1.	Aganiba, B. et al., 2015 (62)	Ghana	0	1	1	1	1	1	1	1	1	1	9	High
2.	Apprey C. et al., 2019 (63)	Ghana	0	1	1	1	1	1	1	1	1	1	9	High
3.	Diendéré, J. et al, 2018 (35)	Burkina Faso	0	1	1	1	1	1	1	1	1	1	9	High
4.	Mkhize X. et al., 2013 (53)	South Africa	0	1	1	1	1	1	1	1	1	1	9	High

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5.	MJ, C. et al, 2012 (69)	Lake Victoria Basin of East Africa	0	1	1	1	1	1	1	1	1	1	9	High
6.	Ijarotimi OS & Keshinro OO, 2004 (65)	Tanzania	0	0	0	1	1	1	1	1	1	1	7	Moderate
7.	Nyaruhucha CNM. et al, 2001 (66)	Tanzania	0	1	1	1	1	1	1	1	1	1	9	High
8.	Olayiwola O.et al, 2006 (47)	Nigria	0	1	1	1	1	1	1	1	1	1	9	High
9.	Rouvray C. et al., 2014 (61)	Central Africa (COR & ROC)	0	1	1	1	1	1	1	1	1	1	9	High
10.	Legesse M. et al., 2019 (34)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
11.	Wondiye K et al., 2019 (35)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
12.	Agbozo, F. et al., 2018 (64)	Ghana	0	1	1	1	1	1	1	1	1	1	9	High
13.	Alao M. et al., 2015 (48)	Nigeria	0	0	0	1	1	1	1	1	1	1	7	Moderate
14.	Tessfamichael, D. et al., 2014 (40)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
15.	Otitoola Oc. Et al., 2015 (57)	South Africa	0	0	0	0	1	1	1	1	1	1	6	Moderate
16.	Pilleron, S. et al., 2015 (58)	Central Africa (COR & ROC)	1	1	1	1	1	1	1	1	1	1	10	High
17.	Esmayel EM. et al, 2013 (42)	Egypt	0	1	1	1	1	1	1	1	1	1	9	High
18.	Andre MB. et al., 2013 (59)	Democratic Republic of Congo	0	1	1	1	1	1	1	1	1	1	9	High
19.	Adebusoye L.A. et al., 2012 (49)	Nigeria	0	1	1	1	1	1	1	1	1	1	9	High
20.	Robb L. et al., 2017 (54)	South Africa	0	1	1	1	1	1	1	1	1	1	9	High
21.	El-Sherbiny N.A. et al., al.2016 (41)	Egypt	0	1	1	1	1	1	1	1	1	1	9	High

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22.	Naidoo I. et al., 2015 (56)	South Africa	0	1	1	1	1	1	1	1	1	1	9	High
23.	Adhana, Z K et al., 2015 (37)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
24.	Adebusoye L. et al., 2019 (50)		0	0	1	1	1	1	1	1	1	1	8	Moderate
25.	Marias ML. et al., 2007 (55)	South Africa	0	1	1	1	1	1	1	1	1	1	9	High
26.	Hailemariam H. et al., 2016 (36)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
27.	Abate et al., 2020 (38)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
28.	Abdu AO et al., 2020 (39)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
29.	Abd Allah et al., 2020 (43)	Egypt	0	0	0	0	1	1	1	1	1	1	6	Moderate
30.	Rasha El-Desouky_2017 (44)	Egypt	0	0	1	1	1	1	1	1	1	1	8	Moderate
31.	Khater MS et al., 2011 (45)	Egypt	0	0	0	0	1	1	1	1	1	1	6	Moderate
32.	F. Mahjoub et al., 2019 (68)	Tunis	0	0	0	1	1	1	1	1	1	1	7	Moderate
33.	Adebusoye, LA. et al., 2018 (52)	Nigeria	0	0	0	0	1	1	1	1	1	1	6	Moderate
34.	Abd-El-Gawad et al., 2014 (51)	Nigeria	0	0	0	1	1	1	1	1	1	1	7	Moderate
35.	Mahfouz, E.M. et al., 2013 (46)	Egypt	0	0	0	1	1	1	1	1	1	1	7	Moderate
36.	Andia A, et al., 2019 (70)	Niger	0	0	0	0	1	1	1	1	1	1	6	Moderate

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Supplementary File

Search items for HINARI/PubMed

#1= "Prevalence"[MeSH] OR "Epidemiology"[MeSH]

#2= "Malnutrition"[MeSH] OR "Obesity"[MeSH]

#3= malnutrition [tiab] OR nutrition*[tiab] OR obesity [tiab]

#4= aged [MeSH] OR elder [MeSH]

#5= old*[tiab] OR aged [tiab] OR elder [tiab] OR geriatric [tiab]

#6= Africa [MeSH] OR Africa [tiab]

#7= Angola[tiab] OR Benin[tiab] OR Botswana[tiab] OR Burkina Faso[tiab] OR Burundi[tiab] OR Cameroon[tiab] OR Cape Verde[tiab] OR Central African Republic[tiab] OR Chad[tiab] OR Comoros[tiab] OR Congo[tiab] OR Côte d'Ivoire[tiab] OR Democratic Republic of the Congo[tiab] OR Djibouti[tiab] OR Equatorial Guinea[tiab] OR Eritrea[tiab] OR Ethiopia[tiab] OR Gabon[tiab] OR Gambia[tiab] OR Ghana[tiab] OR Guinea[tiab] OR Guinea-Bissau[tiab] OR Kenya[tiab] OR Lesotho[tiab] OR Liberia[tiab] OR Madagascar[tiab] OR Malawi[tiab] OR Mali[tiab] OR Mauritania[tiab] OR Mauritius[tiab] OR Mayotte[tiab] OR Mozambique[tiab] OR Namibia[tiab] OR Niger[tiab] OR Nigeria[tiab] OR Réunion[tiab] OR Rwanda[tiab] OR Saint Helena[tiab] OR São Tomé and Príncipe[tiab] OR Senegal[tiab] OR Seychelles[tiab] OR Sierra Leone[tiab] OR Somalia[tiab] OR South Africa[tiab] OR Sudan[tiab] OR Swaziland[tiab] OR Togo[tiab] OR Uganda[tiab] OR United Republic of Tanzania[tiab] OR Zambia[tiab] OR Zimbabwe[tiab] OR Algeria[tiab] OR Egypt[tiab] OR Libya[tiab] OR Morocco[tiab] OR Tunisia[tiab]

#8=#2 OR #3

#9= #4 OR #5

#10= #6 OR #7

#11= #1 AND #8 AND #9 AND #10

#12= Filter: date from 2000/01/01 to 2019/12/10; English language; Humans

Combined search terms

((("Prevalence"[MeSH] OR "Epidemiology"[MeSH]) AND ("Malnutrition"[MeSH] OR "Obesity"[MeSH] OR malnutrition [tiab] OR nutrition*[tiab] OR obesity[tiab]) AND (aged[MeSH] OR elder[MeSH] OR old*[tiab] OR aged[tiab] OR elder[tiab] OR geriatric[tiab]) AND (Africa[MeSH] OR Africa[tiab] OR Angola[tiab] OR Benin[tiab] OR Botswana[tiab] OR Burkina Faso[tiab] OR Burundi[tiab] OR Cameroon[tiab] OR Cape Verde[tiab] OR Central

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African Republic[tiab] OR Chad[tiab] OR Comoros[tiab] OR Congo[tiab] OR Côte d'Ivoire[tiab] OR Democratic Republic of the Congo[tiab] OR Djibouti[tiab] OR Equatorial Guinea[tiab] OR Eritrea[tiab] OR Ethiopia[tiab] OR Gabon[tiab] OR Gambia[tiab] OR Ghana[tiab] OR Guinea[tiab] OR Guinea-Bissau[tiab] OR Kenya[tiab] OR Lesotho[tiab] OR Liberia[tiab] OR Madagascar[tiab] OR Malawi[tiab] OR Mali[tiab] OR Mauritania[tiab] OR Mauritius[tiab] OR Mayotte[tiab] OR Mozambique[tiab] OR Namibia[tiab] OR Niger[tiab] OR Nigeria[tiab] OR Réunion [tiab] OR Rwanda[tiab] OR Saint Helena[tiab] OR São Tomé and Príncipe[tiab] OR Senegal[tiab] OR Seychelles[tiab] OR Sierra Leone[tiab] OR Somalia[tiab] OR South Africa[tiab] OR Sudan[tiab] OR Swaziland[tiab] OR Togo[tiab] OR Uganda[tiab] OR United Republic of Tanzania[tiab] OR Zambia[tiab] OR Zimbabwe[tiab] OR Algeria[tiab] OR Egypt[tiab] OR Libya[tiab] OR Morocco[tiab] OR Tunisia[tiab])))

In case of Google Scholar advanced search for the topic was as follows:

1. With all of the words: old age people, or elderly
2. With at least one of the words: prevalence, epidemiology, malnutrition, undernutrition, overnutrition, obesity, overweight, old age people, older adult, elderly, Africa, Sub-Saharan Africa, Horn of Africa
3. Where my words occur: in the title of the article

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	2
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	3
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	3-4
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	3
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	3
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	3
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	4
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	3-5
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	4
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	4
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	4
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	3
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	4
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	5
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	5
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	5
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	5
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	4
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	N/A

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	5
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	6
Study characteristics	17	Cite each included study and present its characteristics.	6
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	7
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	7-8
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	8
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	8-9
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	9
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	9
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	N/A
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	N/A
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	10
	23b	Discuss any limitations of the evidence included in the review.	12
	23c	Discuss any limitations of the review processes used.	12
	23d	Discuss implications of the results for practice, policy, and future research.	12
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	3
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	3
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	13
Competing interests	26	Declare any competing interests of review authors.	13
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	13

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi:



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Prevalence of malnutrition among old age people in Africa: Systematic Review and Meta-Analysis

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Abstract

Objective: To summarize available evidence and estimate the pooled prevalence of malnutrition among old age people in Africa.

Design: Systematic review and meta-analysis

Setting: Any health care or community research, reporting the prevalence or incidence of malnutrition in Africa from January 2000 to October 2021.

Participants: Old age people, aged above 60 years.

Outcome measures: Malnutrition either under or overnutrition.

Results: A total of 1,442 studies were retrieved based on the search strategy, where only 36 studies (n = 15,266 participants) reported from 11 African countries were included for meta-analysis. The reported prevalence of malnutrition ranges from 2.2 to 77.3% across the continent. Overall, the pooled prevalence of under and overnutrition were 18% (95% CI: 15-22; $I^2 = 98.1$; $p < 0.001$) and 33% (95% CI: 22-44; $p < 0.001$), respectively.

Conclusion: The prevalence of malnutrition in African old age people is high and differs by setting, assessment tool, and country of residence. Hence, due attention to geriatric nutrition is mandatory, and the need for a valid, reliable, and simple screening tool should be thought of.

Strengths and limitations of this study

- This provides a comprehensive estimate of the prevalence of malnutrition among old

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age people in Africa, including more recent studies conducted in the continent.

- Looked at the prevalence of malnutrition in the elderly using any of the validated screening tools.
- Only HINARI/PubMed and Google Scholars were retrieved.
- Limited to include only English literature.

Key words: Africa; Malnutrition; Old age people; Prevalence; Meta-analysis; Systematic-review

Introduction

Background

The global demographic structure of the population is changing dramatically, notably in Africa, due to several social and economic milestones (1). These resulted in expanding population, improved life expectancy, and increased population size of older ages (1,2). The absolute number of old age people (60 years and above (3)) is expected to grow more rapidly in the coming decades than in any other part of the world (1).

The world is also in nutrition transition, a phenomenon where the changes in the foods and beverages consumed along with the reduced physical activity. In Africa, the risk of malnutrition among older people is a major challenge to the health care system and needs special and urgent attention. The usual food production practices, marketing style, and living standards have all contributed to an increase in the use of low-cost, packaged products that are high in fat, energy, and salt and yet low in nutritional quality. More importantly, poor infrastructure and limited resources, combined with conflict and poor access to healthcare services, are factors that contribute to the overwhelming levels of malnutrition and food insecurity on the continent (4). The lack of policy direction for old age people, combined with prevalent chronic illnesses, adds a significant burden to the growing burden of malnutrition (5,6). Due to these and other aggravating factors, the government's commitment to ending all forms of malnutrition by 2030 (7) will be difficult to achieve.

While any age group may suffer from malnutrition, it is most common among old age people due to the changes in physiological, psychosocial, and health characteristics of individuals in this age group (8). Furthermore, malnutrition has serious and life-threatening consequences that are known

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to be the main causes of increased morbidity and mortality among old age people (9). Besides, complications of malnutrition, such as osteoarthritis, osteoporosis, diabetes, cardiovascular disease, and hypertension, inflict a significant social and economic burden on them (10).

Previous studies reported that the prevalence of malnutrition among old age people varies between 1.1% and 72.2% in different settings (11–15), where some of the variations could be attributed to differences in the measurement tools, study settings, and demographic groups that have been studied (16,17). Moreover, none of the previous reviews targeted the old age people in Africa (18–21), where the root causes of malnutrition are expected to be large. However, having concrete evidence on the burden of malnutrition using a more rigorous and systematic manner than individual pocket studies is the first step toward enhancing the elderly's interventions and policy direction. Context-specific evidence will provide better inform decision-makers in making informed decisions on addressing malnutrition among old age people for a better quality of life.

Objective

To estimate the prevalence of malnutrition among old age people in Africa according to the assessment tools, study setting, and regional location.

Methods

Search strategies

Before the beginning of this systematic review, we conducted a systematic search of review papers from the Cochrane Library, the International Prospective Register of Systematic Reviews (PROSPERO), and the Joanna Briggs Institute (JBI), and no review had been conducted on malnutrition among old age people in Africa. Yet, review protocol was not registered. Relevant publications were searched systematically from HINARI, PubMed, and Google Scholar sources from 01 January 2000 to 16 October 2021. We employed the search through relevant combinations of MeSH terms and related keywords, such as "malnutrition", "obesity", "overweight", "old age people", "older adults", "elderly", and "Africa" as indicated in **Supplementary File**. Lists of references from relevant studies were also manual-searched. Countries or territories included in Africa were defined according to the United Nations (UN) classification (22). Studies were imported and checked for duplicates in Mendeley Desktop (Version 1.19.3).

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Eligibility criteria

Inclusion criteria

Based on the condition, context, and population (CoCoPop) framework (23) we included research publications in English language from January 1, 2000 to October 16, 2021 that reported the prevalence of malnutrition among old age persons aged 60 years and up in African countries. Intending to get a relatively up-to-date prevalence estimate and obtain a comparable age and malnutrition definition, we restricted studies conducted starting in 2000. All studies that reported the prevalence and/or incidence of malnutrition among old age people were considered regardless of the study design. When multiple studies reported the prevalence of malnutrition based on the same study, the one with the larger sample size was included.

Exclusion criteria

Qualitative studies, study protocols, abstracts, opinions, commentaries, case reports, review papers, and studies on African-Americans who do not live on the African continent were all excluded from the review. Publications reporting malnutrition in post-intervention were excluded. Duplicated articles and articles without access to the full text were removed from the review.

Outcome measuring tools

Studies that measured malnutrition with any standard and validated tools, including questionnaires such as Patient-Generated Subjective Global Assessment (PG-SGA), Mini Nutritional Assessment (MNA), Subjective Global Assessment tools (SGA), or Nutritional Screening Checklist (NCL), anthropometric [mid-upper arm circumference (MUAC), calf-circumference (CC), body mass index (BMI), etc.], or biochemical indices (serum haemoglobin, albumin, etc.) were considered. We used the reported prevalence of malnutrition or undernutrition based on the standard definition set for each nutritional assessment method. When the entire text of an article could not be obtained or if there was missing information, the corresponding author(s) was contacted through email. Requested full-text papers that were not submitted until October 31, 2021, and the study was discarded.

Study selection and quality assessment

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A two-step process was used to select the relevant publications. First, the scan of the titles and abstracts was conducted for the inclusion of references that meet the eligibility criteria. The entire text was retrieved when any doubts existed about the relevance of a study. Second, the full texts were then used to make a judgement on whether or not the study should be included in the review. The two authors work individually on all components of the selection process. Disagreements were settled through conversation.

Selected studies were evaluated using a ten-item rating checklist designed for research reporting prevalence (24), where better quality articles have a higher score as indicated in **Supplementary Table 1**.

Data extraction

The two authors separately extracted data from the full-text articles using a pre-defined standard format produced in Excel. The extract comprised the following information: general (authors, year of publication, and country), participants (age, gender, and morbidity status), and study details (study design, study setting, sample size, measurement tool used, and outcome measures). Morbidity was coded as one or multi when a single or multiple underlying health conditions were an inclusion criterion.

Statistical analysis

The pre-coded extracted data in excel format was exported to STATA software version 14 (25) for descriptive and meta-analysis. For each study, the unadjusted prevalence of malnutrition and standard errors were calculated, and a logarithmic (log) transformation for proportional data was used to reduce variance and obtain the pooled prevalence as an effect size (ES). Considering the level of methodological heterogeneity and variance within the study, a random effect model of the DerSimonian and Laird (26) method was used to conduct the meta-analysis.

The heterogeneity between studies was assessed using a forest plot, Cochran's Q test, and I-square (I^2) test. The I^2 values of 25, 50, and 75% with a *p-value* less than 0.10 represent low, medium, and high heterogeneity, respectively (27). Publication bias was checked graphically in a funnel plot supplemented by Egger's statistics (28). We also applied a subgroup analysis, a sensitivity analysis, and meta-regression to identify the potential sources of heterogeneity. A statistically significant effect size estimate has a P-value of less than 0.05.

Patient and public involvement

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This study did not involve any patients or the general public.

Results

Search results

A total of 1,442 articles were obtained through searches, of which 206 were removed due to duplicates. With the title and abstract screening, 971 articles were removed depending on the focus of the review (older age in the African continent), relevant nutritional assessment method, and language (other than English). Furthermore, 237 articles were retrieved during the full-text screening. Finally, 37 and 36 articles were selected for qualitative (systematic review) and quantitative analysis (meta-analysis), respectively. The comprehensive selection process presented using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (29) (**Figure 1).**

Characteristics of included studies

A total of 37 studies reported the prevalence of malnutrition among old age people. While 34 studies (92%) were published in the last decades between 2011 and 2020, the remaining 8% were issued between 2001 and 2010. All the included studies were reported from 11 African countries, where more than half (65%, n = 24) were from four countries: Ethiopia, Egypt, Nigeria, and South Africa.

Seven studies (18.9%) were from Ethiopia (30–36) and six studies (16.2%) each from Egypt (37–42) and Nigeria (43–48). While five studies (13.5%) were from South Africa (49–53), four studies (10.8%) were from the Central African Republic (CAR) and the Republic of Congo (54–57) three studies (8.1%) from Ghana (58–60), and two studies (6.06%) were from Tanzania (61,62).

Except for three studies (two cohorts (31,47) and one case-control (44)), all other identified studies were cross-sectional study designs. Malnutrition prevalence was reported in four studies as a comparison between institutional versus noninstitutional (44,61), high versus low economic classes (50), and with and without dementia (57). However, one study from Tanzania examined the nutritional status after a post-supplemental food program (63) and was excluded from the meta-analysis. Majority of the studies (n = 22) were conducted in community settings, while 11 studies were conducted among institutionalized elders.

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Moreover, a total of 15,266 study participants were found, of whom more than half (58%) were females. The selected studies had sample sizes ranging from 40 (64) to 2,219 (37) people. Regarding the nutritional assessment tools used, more than half of the studies reported malnutrition based on either the full-or short-form of the Mini Nutritional Assessment tool (MNA-FF: n = 15, MNA-SF: n = 4). While 18 studies employed various anthropometric dimensions such as BMI, MUAC, and CC separately or in combinations to assess malnutrition. One study used an interviewer- administered Nutritional Screening Checklist (NCL)(38) and another study utilized a combination of MNA and the geriatric nutrition risk index (GNRI)(64). Two-thirds (n = 23) of the studies reported single (n = 8) or multiple (n = 15) underlying clinical conditions such as fracture, heart disease, dementia, Parkinson's disease, or cancer (**Supplementary Table 2**).

Quality of the studies

According to the quality assessment checklist, studies were either in moderate (6-8 points) (39,40,64,65,41,42,44,46–48,53,61) or high-quality groups (9 or 10 points) (30,31,45,49–52,54,55,57–59,32,60,62,66,33–38,43). Study bias was mostly exacerbated as a result of convenience sampling and a low response rate among study participants. In almost all studies, the possibility of external validity was low as the target population was not nationally representative of the country. However, the quality of two studies (56,63) was not assessed and hence not processed for quantitative data analysis. The first was (56) due to duplication of results from previous data (57) and the latter (63) was due to the outcome variable being measured after an intervention (**Supplementary Table 3**).

Prevalence of malnutrition

The prevalence of malnutrition varied highly across the studies conducted in different countries, assessment tools used, and healthcare settings. We presented an overview of the findings from various studies as under and overnutrition using two assessment tools in the section below.

Prevalence of undernutrition

Seventeen studies from hospitals (n = 5), the communities (n = 10), a long-term care center (n = 1), and a daycare center (n = 1), utilized body mass index (BMI) as a nutritional assessment tool.

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As low as zero (53) and 4.0% (49) of research participants respectively from the elderly daycare center and hospital settings in South Africa were classified as undernourished (BMI ≤ 18.5 kg/m^2). Furthermore, the highest prevalence of undernutrition (25.6%) was reported in a Tanzanian hospital (62). From the community settings, undernutrition was reported at 9.9% in Ghana (59) and 26.4% in the East Africa (66).

On the other hand, 20 studies from hospitals (n = 7), communities (n = 11), and long-term care centers (n = 2), utilized mini nutritional assessment (MNA) as a nutritional assessment tool. The outcome was reported as at risk of undernutrition, undernutrition, or normal status. The results showed that old age people in the community were more victims of undernutrition compared to hospitals and long-term care centers. The prevalence ranged from 5.7% in Niger (65) to 56% in Egypt (38) and 18% in Egypt (38) to 58.5% in Niger (65), respectively. While in Nigerian hospitals, 2.24% (48) to 40.46% (47) and 11.8% (45) to 52.7% (47) of old age people were undernourished or at risk of malnutrition, respectively. Besides, 7.3% in South Africa (50) and 10.8% in Egypt (41) were undernourished using MNA from the long-term care center. Similarly, 10.9% of community study participants in Egypt (37) and 77.3% in Ethiopia (33) were undernourished, while none from Ethiopia(33) and 41.9% from Egypt (37) were at risk of undernutrition using MNA-SF. Moreover, 4.57% of hospital study participants in Nigeria (46) were undernourished, while 95.43% were either normal or at risk of undernutrition.

Prevalence of overnutrition

Overnutrition was reported in 17 studies that used BMI as a nutritional assessment tool. Accordingly, 77% of the elderly people studied in a South African hospital (49) were overweight or obese. There were none from Tanzania (62) in a comparable situation. While 15.3% of study participants in East Africa (66) and 42.1% in Ghana (59) were affected by overweight or obesity in the community settings.

Meta-analysis

Adequate data were available to conduct pooled malnutrition prevalence estimates for two nutrition screening tools: under nutrition from 36 studies (BMI = 16 and MNA = 20), and overnutrition from 16 studies using only BMI. Since the full-and short-form of MNA and NCL

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tools are similar in the interpretation of the outcome variable, they were treated as similar tools. The random pooled prevalence of under and overnutrition in Africa were 18% (95% CI: 15-22; p < 0. 001) and 33% (95% CI: 22-44; p < 0. 001), respectively **(Figure 2 and 3)**.

A high level of statistical heterogeneity was observed between studies (heterogeneity of Cochran Q statistics = 1828.98 with *df* = 35 and *p-value* < 0.001; $I^2 = 98.1\%$; *Tau-square* ($\tau^2 = 0.01$).

Moreover, a minimal publication bias has been noted in the published studies. Studies with lower standard error and larger sample size might be included, as indicated in of the funnel and Egger's publication plots (**Figure 4**). Smaller studies with larger standard errors plotted on the right of the funnel plot tend to have both smaller and larger odds ratios. This may indicate the existence of bias due to small-study effects. Moreover, the Egger plot shows that the data near the origin are unsystematically elevated, where the confidence interval (CI) does not include zero, indicating asymmetry in the funnel plot and evidence of publication bias.

Furthermore, Egger's regression asymmetry test value was 3.50 (p = 0.008), suggesting publication bias. Here, the positive coefficient indicates that small studies overestimate the effect size. However, no study appeared to have a significant impact on the summary effect size in the sensitivity analysis, strongly suggesting that the effect estimate is consistent across groups and potentially giving a reliable estimate of the outcome. In addition, the Trim and Fill analysis was performed to examine the potential impact of publication bias on the interpretation of the data. The method adds 12 "missing" studies to the dataset. However, the correction for publication bias does not change the overall interpretation of the findings.

Subgroup analysis and meta-regression

Subgroup analysis and meta-regression were performed for the study regions, settings, assessment tools, age group, and presence of multimorbidity using the random-effects inverse-variance model with the DerSimonian-Laird estimate. In general, a relatively high prevalence of malnutrition was observed in central Africa (3.76%; 95% CI: 3.22-4.39) as compared to other regions of the continent. Similarly, the community elderly (3.12%; 95% CI: 2.66-3.67), older age groups (3.54%; 95% CI: 2.32–5.41), and those with no reported clinical condition (3.16%; 95CI: 2.62–3.80) had a higher malnutrition burden than their counterparts. The prevalence estimates

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had a statistically significant difference among all subgroups except for the assessment tools used (p = 0.430) (Table 4).

Table 4: Subgroup analysis for the prevalence of malnutrition among old age people, 2021

Subgroup	No of studies included	Prevalence with 95%CI	Cochran's Q statistics for heterogeneity	P value	I^2
Regions					
Northern Africa	8	2.82[2.72-2.92]	3.93	0.788	0.0%
Western Africa	10	2.46[1.95-3.11]	263.06	< 0.001	96.6%
Central Africa	3	3.76[3.22-4.39]	2.08	0.354	3.7%
Eastern Africa	10	3.46[2.29-5.22]	312.49	< 0.001	97.1%
Southern Africa	5	1.79[1.41-2.26]	60.54	< 0.001	93.4%
Between groups			42.08	< 0.001	
Study settings					
Community	21	3.12[2.66-3.67]	829.56	< 0.001	97.6%
Hospital	11	2.12[1.67-2.70]	132.88	< 0.001	92.5%
Long-Term Care	3	2.34[2.08-2.63]	0.31	0.859	0.0%
Elderly day care center	1	1.04[0.87-1.24]	0.00	-	-
Between groups			86.79	< 0.001	
Assessment Tools					
BMI	16	2.98[2.37-3.75]	213.68	< 0.001	93.0%
NCL	1	5.74[0.71-46.74]	-	-	-
MNA	15	2.45[2.11-2.85]	264.09	< 0.001	94.7%
MNA-SF	4	2.45[1.96-3.06]	136.91	< 0.001	97.8%
Between groups			2.76	0.430	
Age group					
≥ 60	26	2.54[2.17-2.98]	1102.79	< 0.001	97.7%
≥ 65	10	3.54[2.32-5.41]	361.43	< 0.001	97.5%
Between groups			2.06	< 0.001	
Morbidity					
One	7	1.66[1.46-1.89]	91.58	< 0.001	93.4%
Multi	16	2.84[2.50-3.22]	205.87	< 0.001	92.7%
Not reported	13	3.16[2.62-3.80]	36.06	< 0.001	66.7%
Between groups			16.44	< 0.001	
Overall	**37**	**2.77(2.43-3.16)**	**1573.48**	**< 0.001**	**97.7%**

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However, only the study setting was statistically significant (adjusted R-squared = 39.2%, p = 0.049) predicting the prevalence of malnutrition in the meta-regression analysis.

Discussion

We conducted this review paper with the intention of estimating the pooled prevalence of malnutrition disaggregated by relevant factors such as study setting, screening tool, and so on. According to the findings, the prevalence of malnutrition ranges from 4.0% (49) to 77.3% (33) across countries, depending on the assessment tools used and the study settings. These prevalence figures are consistent with previous European estimates of 3.8 to 67.4% (19,21) among older people despite differences in the settings. In contrast, the highest risk of malnutrition was observed in the community-dwelling old age people assessed by the MNA (both full and short forms) than in other studies; 7.5% to 77.3% versus 5% (18) to 19% (67). The variation may be due to the economic variables of the countries.

Overall undernutrition was 17.1% in the African continent using BMI, which is comparable to European countries, 3.8-18.2% (21). While 27% of the study participants in the continent were overweight or obese. Furthermore, many studies (n=14) in our review used BMI as a nutritional assessment tool for old age people. Similar studies in Africa still preferred to use BMI for old age people where 19.9% and 30% of subjects were undernourished, overweight or obese, respectively (68).

However, the BMI cutoff point used to identify underweight can have a significant impact on prevalence estimates, which tend to misclassify the elderly with physical spinal deformities and other factors that can affect height measurements (69). Due to such measurement errors encountered, the assessment of nutritional status in the old age people become a challenging task, which warrants more comprehensive and multidimensional assessments including clinical examination, anthropometric measures, laboratory tests, dietary surveys, and social aspects (70,71). This means one assessment tool may not be suitable for all settings, which need contextualization (18). Though more than 22 best-validated nutrition screening tools were identified for older adults (17,19), there has long been a lack of consensus regarding the criteria needed to make a diagnosis of malnutrition.

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On the other hand, the meta-analysis revealed a pooled prevalence of malnutrition of 18% (95% CI: 15-22; $p < 0.001$). This is in the range of the ones reported from the world, ranging from 0.8 to 24.6% (72) depending on the study regions, settings, and assessment tools used. The increasing number of an ageing population in Africa and nutritional transitions might have contributed to a higher burden of malnutrition. Our estimate is also consistent with the pooled prevalence of Indian studies (18.29%; 95% CI: 15.24-21.57) (73). However, the result is slightly higher than the prevalence of malnutrition reported from the Central Demographic Republic of Congo and Nigeria (14.5%; 95% CI: 0.0-40.4%) (72). This disagreement could be explained by the use of various assessment procedures and the merging of pooled estimates from two countries where the burden of malnutrition might be higher.

Surprisingly, the current prevalence estimate is lower than the reviews reported from European older persons (48.4%; 95% CI: 41.5-51.8) (19). The disparity could be attributed to economic factors, health literacy, and healthcare access, and the majority of the previous review data came from hospital settings, implying that individuals with higher health risks were included.

Still, there was significantly high heterogeneity between the studies ($Q = 1873.99$; $df = 36$; $p < 0.001$; $T^2 = 0.01$; $I^2 = 98.1\%$). Sensitivity analysis did not improve this heterogeneity, which is consistent with other previous findings (18,72). Because of the highly complex physiological, social, and temporal nature of malnutrition, which differs from person to person, as well as bias introduced by research design, analysis of the pooled prevalence of malnutrition is difficult to account for all sources of variation (74).

In case of subgroup analysis, the frequency of malnutrition showed a great variation depending on the study regions, study settings, age groups, and clinical conditions of study participants. Malnutrition was found to be prevalent in central Africa (3.76%), in older age groups (3.54%), with no documented clinical condition (3.16%), and in community-dwellers (3.12%). The prevalence was statistically significant in all subgroups except between the assessment tools used ($p = 0.430$). These findings are consistent with the reviews from India (73) and the rest of the world (72). While the meta-regression identified the study setting as a statistically significant (adjusted $R^2 = 39.20\%$, $p = 0.049$) predictor of malnutrition prevalence. This is, however, different from the reviews conducted in India (73) and the world (72) where the study region ($R^2 = 27.1\%$, $p = 0.026$) and assessment tools ($R^2 = 76.47\%$, $p<0.001$) were statistically

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significant predictors of the malnutrition prevalence, respectively. The variation might be due to the difference in assessment tools used between the studies.

Strength and Limitation

To our knowledge, this is the first systemic review and meta-analysis that provides a comprehensive estimate of the prevalence of malnutrition among old age people in Africa, including more recent studies conducted in the continent. Even though this review has its strengths, it is not without limitations. Due to the inherent heterogeneity among the included studies (I^2 = 98.08%), the interpretation of the current study needs precaution. Being limited to including only English literature and not including research published in books, magazines, and others might limit the generalizability of the estimate.

Conclusion

This systematic review and meta-analysis showed high malnutrition prevalence among old age people which warrants a targeted policy intervention. The prevalence varied greatly depending on the study setting, the tool used, and the country of residence. Due attention to geriatric nutrition is mandatory, and to better monitor and evaluate, the use of a more reliable, valid, and standardized malnutrition assessment tool is strongly recommended.

List of abbreviations

BMI: Body Mass Index; CC: Calf Circumference; HINARI: Health InterNetwork Access to Research Initiative; MeSH: Medical Subject Headings; MNA: Mini Nutritional Assessment

Declarations

Ethical approval and formal consent

Not applicable

Consent for publication

Not applicable

Availability of data and materials

All data generated or analyzed during this study are included in this published article and its supplementary information files, tables, and figures.

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Competing interests

The authors declare that they have no competing interests

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The study was created and designed by AM under the supervision and approval of NF. They are both responsible for all parts of this work. The final manuscript was read and approved by both writers.

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Figure legends

Figure 1: PRISMA flow-chart of studies screened and reviewed.

Figure 2: Forest plot displaying the pooled prevalence of undernutrition among old age people in Africa.

Figure 3: Forest plot displaying the pooled prevalence of overnutrition among old age people in Africa.

Figure 4*:* Probability of publication bias for included studies (a) funnel plot and (b) Egger's publication bias plot.

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Figure Legends

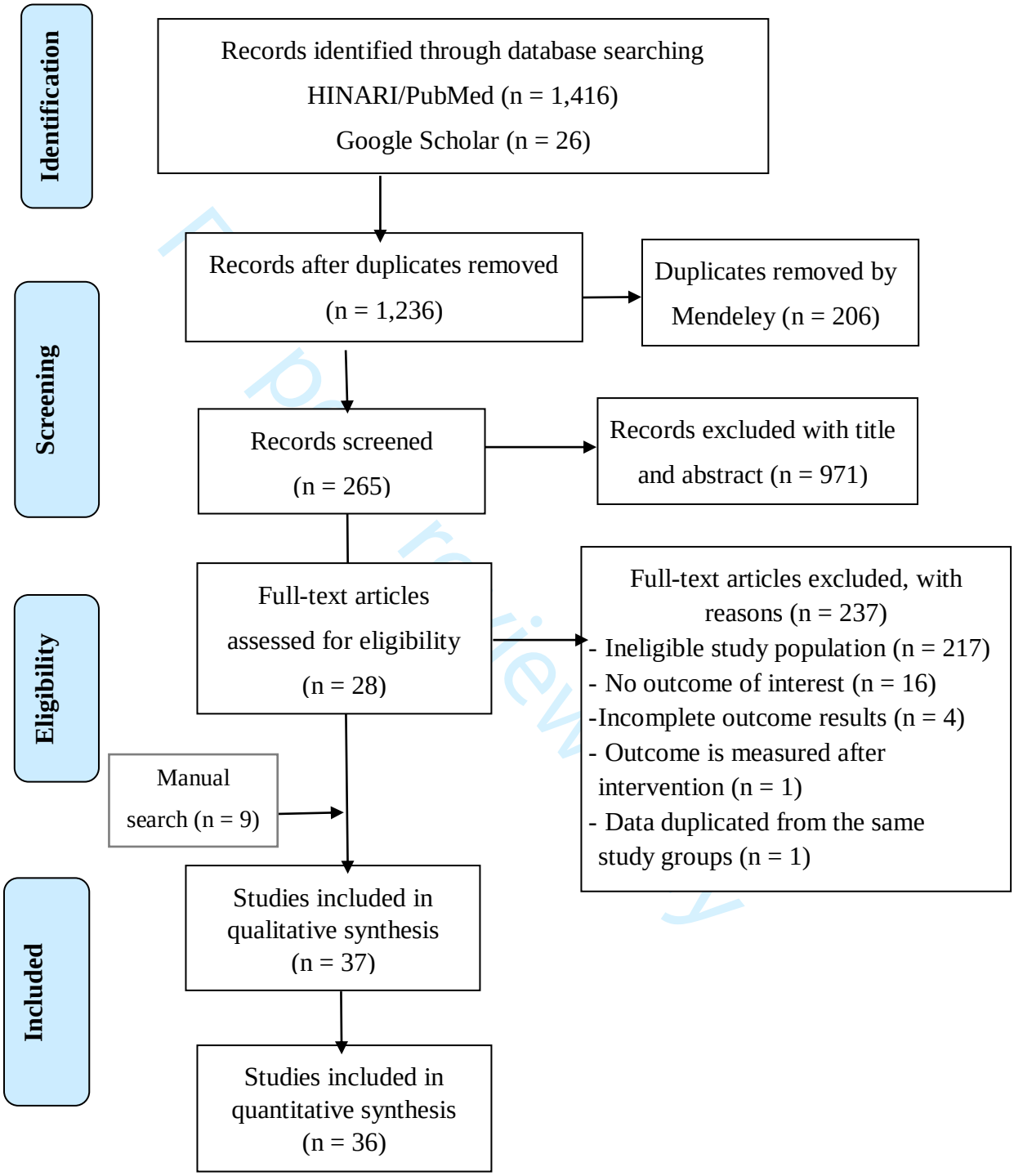


Figure 1: PRISMA flow-chart of studies screened and reviewed.

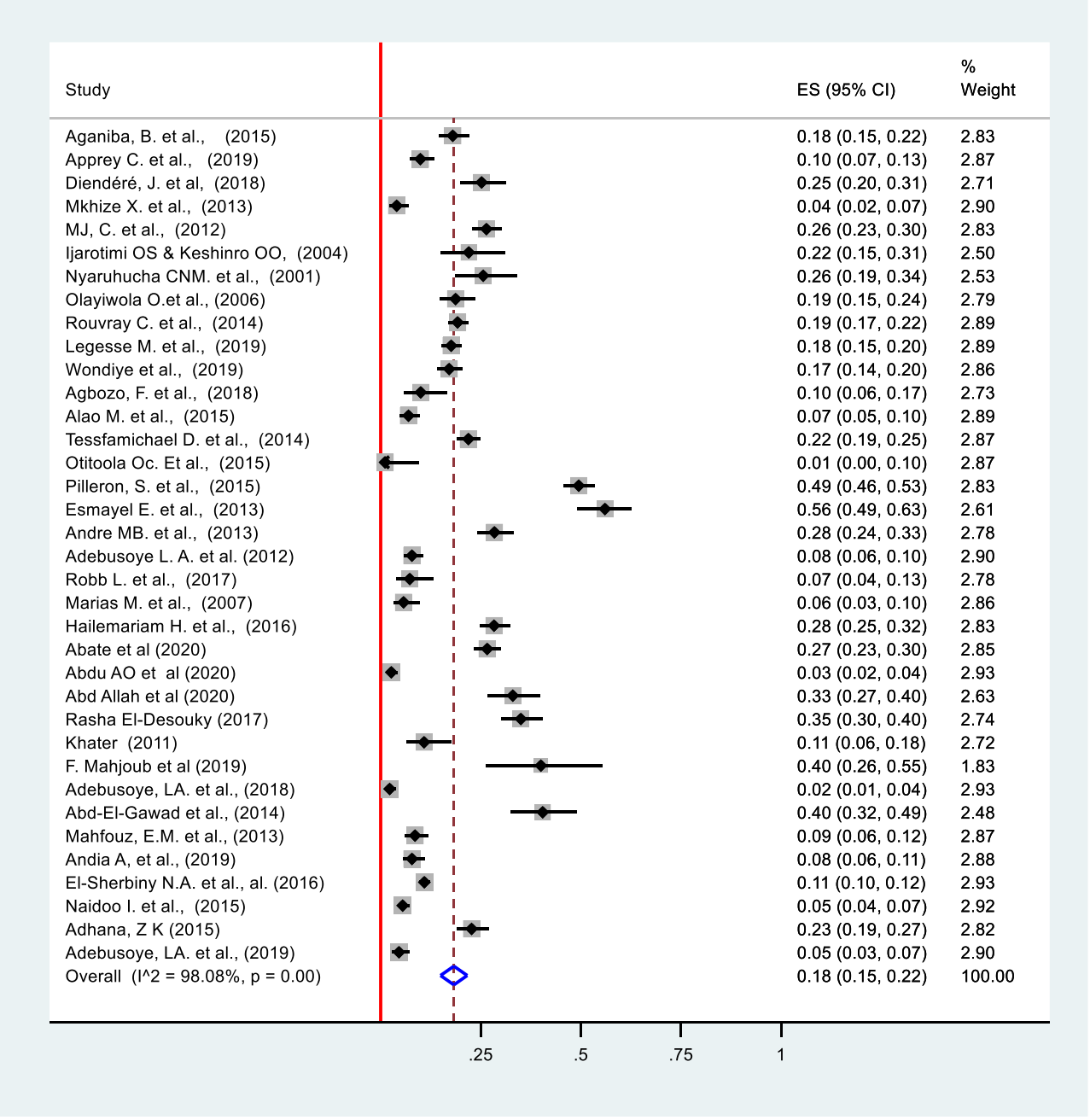


Figure 1: Forest plot displaying the pooled prevalence of undernutrition among old age people in Africa.

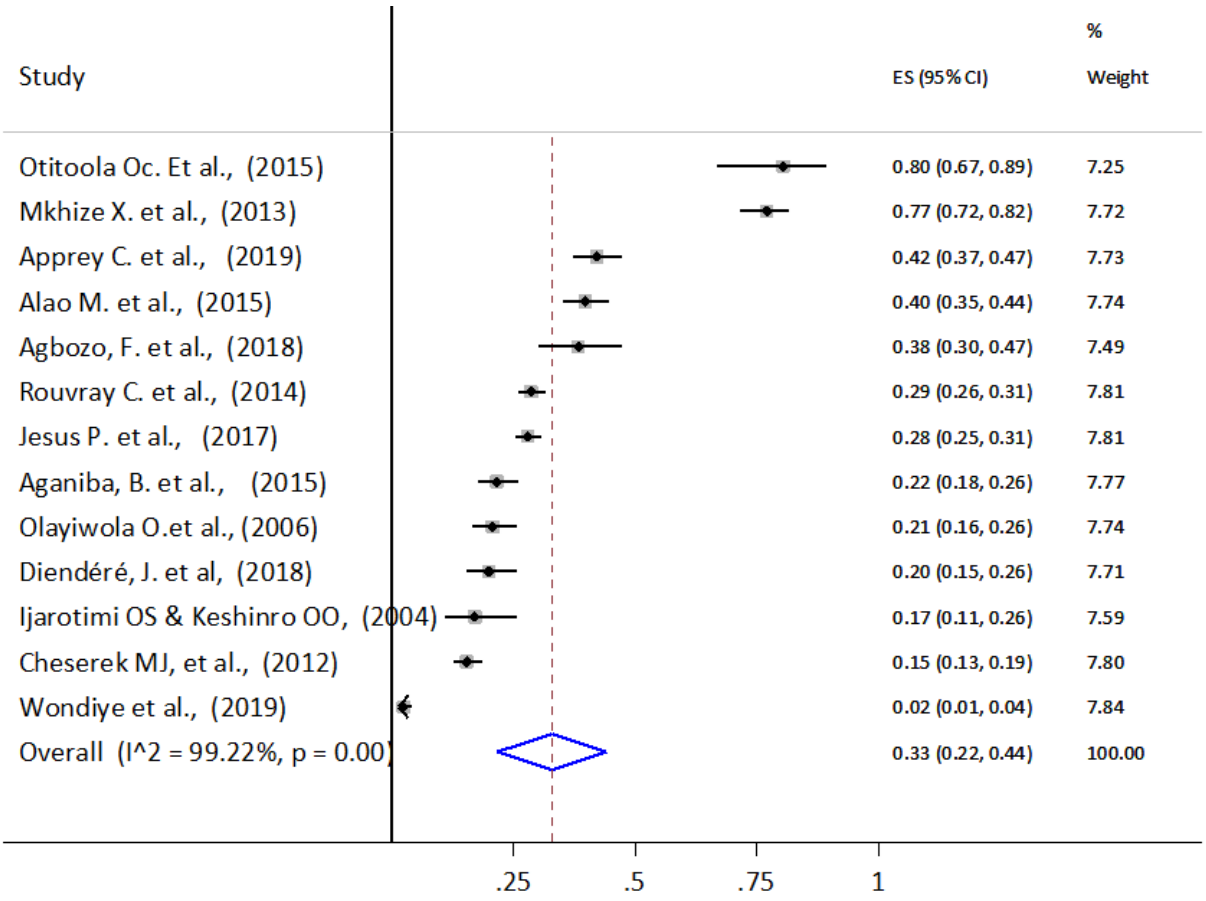
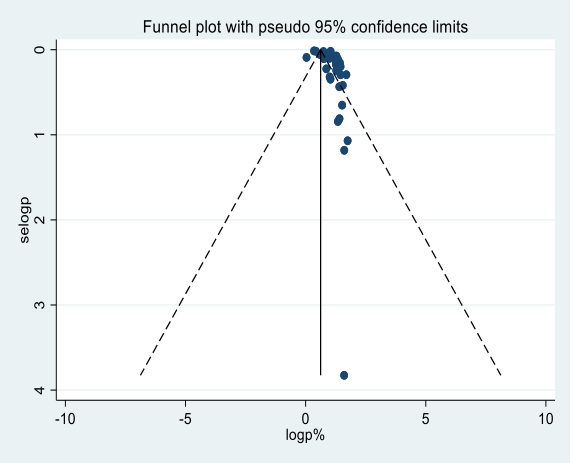


Figure 3: Forest plot displaying the pooled prevalence of overnutrition among old age people in Africa.

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a.



b.

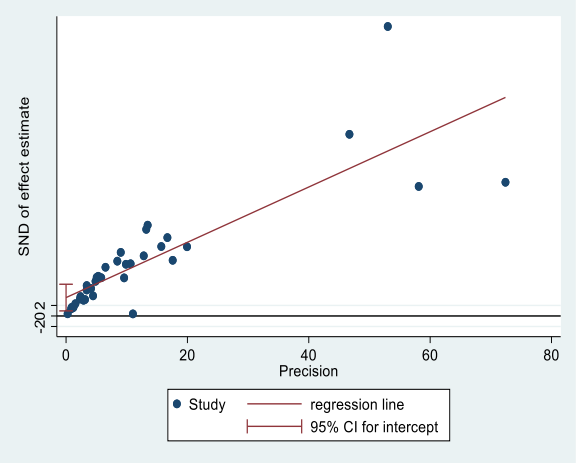


Figure *4:* Probability of publication bias for included studies (a) funnel plot and (b) Egger's publication bias plot.

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Supplementary File

Search items for HINARI/PubMed

#1= "Prevalence"[MeSH] OR "Epidemiology"[MeSH]

#2= "Malnutrition"[MeSH] OR "Obesity"[MeSH]

#3= malnutrition [tiab] OR nutrition*[tiab] OR obesity [tiab]

#4= aged [MeSH] OR elder [MeSH]

#5= old*[tiab] OR aged [tiab] OR elder [tiab] OR geriatric [tiab]

#6= Africa [MeSH] OR Africa [tiab]

#7= Angola[tiab] OR Benin[tiab] OR Botswana[tiab] OR Burkina Faso[tiab] OR Burundi[tiab] OR Cameroon[tiab] OR Cape Verde[tiab] OR Central African Republic[tiab] OR Chad[tiab] OR Comoros[tiab] OR Congo[tiab] OR Côte d'Ivoire[tiab] OR Democratic Republic of the Congo[tiab] OR Djibouti[tiab] OR Equatorial Guinea[tiab] OR Eritrea[tiab] OR Ethiopia[tiab] OR Gabon[tiab] OR Gambia[tiab] OR Ghana[tiab] OR Guinea[tiab] OR Guinea-Bissau[tiab] OR Kenya[tiab] OR Lesotho[tiab] OR Liberia[tiab] OR Madagascar[tiab] OR Malawi[tiab] OR Mali[tiab] OR Mauritania[tiab] OR Mauritius[tiab] OR Mayotte[tiab] OR Mozambique[tiab] OR Namibia[tiab] OR Niger[tiab] OR Nigeria[tiab] OR Réunion[tiab] OR Rwanda[tiab] OR Saint Helena[tiab] OR São Tomé and Príncipe[tiab] OR Senegal[tiab] OR Seychelles[tiab] OR Sierra Leone[tiab] OR Somalia[tiab] OR South Africa[tiab] OR Sudan[tiab] OR Swaziland[tiab] OR Togo[tiab] OR Uganda[tiab] OR United Republic of Tanzania[tiab] OR Zambia[tiab] OR Zimbabwe[tiab] OR Algeria[tiab] OR Egypt[tiab] OR Libya[tiab] OR Morocco[tiab] OR Tunisia[tiab]

#8=#2 OR #3

#9= #4 OR #5

#10= #6 OR #7

#11= #1 AND #8 AND #9 AND #10

#12= Filter: date from 2000/01/01 to 2019/12/10; English language; Humans

Combined search terms

((("Prevalence"[MeSH] OR "Epidemiology"[MeSH]) AND ("Malnutrition"[MeSH] OR "Obesity"[MeSH] OR malnutrition [tiab] OR nutrition*[tiab] OR obesity[tiab]) AND (aged[MeSH] OR elder[MeSH] OR old*[tiab] OR aged[tiab] OR elder[tiab] OR geriatric[tiab]) AND (Africa[MeSH] OR Africa[tiab] OR Angola[tiab] OR Benin[tiab] OR Botswana[tiab] OR Burkina Faso[tiab] OR Burundi[tiab] OR Cameroon[tiab] OR Cape Verde[tiab] OR Central

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African Republic[tiab] OR Chad[tiab] OR Comoros[tiab] OR Congo[tiab] OR Côte d'Ivoire[tiab] OR Democratic Republic of the Congo[tiab] OR Djibouti[tiab] OR Equatorial Guinea[tiab] OR Eritrea[tiab] OR Ethiopia[tiab] OR Gabon[tiab] OR Gambia[tiab] OR Ghana[tiab] OR Guinea[tiab] OR Guinea-Bissau[tiab] OR Kenya[tiab] OR Lesotho[tiab] OR Liberia[tiab] OR Madagascar[tiab] OR Malawi[tiab] OR Mali[tiab] OR Mauritania[tiab] OR Mauritius[tiab] OR Mayotte[tiab] OR Mozambique[tiab] OR Namibia[tiab] OR Niger[tiab] OR Nigeria[tiab] OR Réunion [tiab] OR Rwanda[tiab] OR Saint Helena[tiab] OR São Tomé and Príncipe[tiab] OR Senegal[tiab] OR Seychelles[tiab] OR Sierra Leone[tiab] OR Somalia[tiab] OR South Africa[tiab] OR Sudan[tiab] OR Swaziland[tiab] OR Togo[tiab] OR Uganda[tiab] OR United Republic of Tanzania[tiab] OR Zambia[tiab] OR Zimbabwe[tiab] OR Algeria[tiab] OR Egypt[tiab] OR Libya[tiab] OR Morocco[tiab] OR Tunisia[tiab])))

In case of Google Scholar advanced search for the topic was as follows:

1. With all of the words: old age people, or elderly
2. With at least one of the words: prevalence, epidemiology, malnutrition, undernutrition, overnutrition, obesity, overweight, old age people, older adult, elderly, Africa, Sub-Saharan Africa, Horn of Africa
3. Where my words occur: in the title of the article

Supplementary Table 1: Quality appraisal checklist for prevalence studies

Items		**Quality score**
	External validity Criteria	
1.	Was the study's target population a close representation of the national population in relation to relevant variables?	(1 point)
2.	Was the sampling frame a true or close representation of the target population?	(1 point)
3.	Was some form of random selection used to select the sample, OR was census undertaken?	(1 point)
4.	Was the likelihood of nonresponse bias minimal?	(1 point)
		Total (4 points)
	Internal validity Criteria	
5.	Were data collected directly from the subjects (as opposed to a proxy)?	(1 point)
6.	Was an acceptable case definition used in the study?	(1 point)
7.	Was the study instrument that measured the parameter of interest shown to have validity and reliability?	(1 point)
8.	Was the same mode of data collection used for all subjects?	(1 point)

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9.	Was the length of the shortest prevalence period for the parameter of interest appropriate?	(1 point)
10.	Were the numerator(s) and denominator(s) for the parameter of interest appropriate?	(1 point)
		Total (6 points)
	Overall total	10 points

Methodological quality

High quality: score > 8

Moderate quality: score 6-8

Low quality: score 0-5

(Source: Hoy et al.(24))

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Supplementary Table 2. Characteristics of Selected Studies Using Different Nutritional Assessment Tools

S/no	Authors	Country	Study design	Study setting	Sample size	Age group	Gender		Assessment tool used	Nutritional status (outcome variable) in %				Multi-morbidity
							Male	Female		Under-nutrition	At risk of malnutrition	Normal-weight/ Nutrition	Overweight/ obesity	
1.	Aganiba, B. et al., 2015 (62)	Ghana	C/S	Community	400	≥ 65	177	223	BMI	18.0		60.5	21.5	Not reported
2.	Apprey C. et al., 2019 (63)	Ghana	C/S	Community	375	≥ 60	127	248	BMI	9.9		48	30.9/11.2	Multi
3.	Diendéré, J. et al, 2018 (35)	Burkina Faso	Cohort	Hospital	222	≥ 60	121	101	BMI	25.2		55.0	14.9/5.0	One
4.	Mkhize X. et al., 2013 (53)	South Africa	C/S	Hospital	270	≥ 60	46	224	BMI	4.0		19.0	24.0/53.0	One
5.	MJ, C. et al, 2012 (69)	Lake Victoria Basin of East Africa	C/S	Community	537	≥ 60	227	310	BMI	26.4		58.3	10.8/4.5	Not reported
6.	Ijarotimi & Keshinro, 2004 (65)	Tanzania	C/S	Hospital	100	≥ 60	51	49	BMI	22.0		61.0	17.0	Not reported
7.	Nyaruhucha CNM. et al, 2001 (66)	Tanzania	C/S	Hospital	121	≥ 60	67	54	BMI	25.6		74.4	0	Not reported
8.	Olayiwola O.et al, 2006 (47)	Nigeria	C/S	Hospital	295	≥ 60	153	142	BMI	18.7		60.7	15.7/4.9	Not reported
9.	Jésus p. et al., 2017 (60)	CAR & ROC	C/S	Community	990	≥ 65	405	585	BMI	19.2		52.9	19.1/8.8	Multi
10.	Rouvray C. et al., 2014 (61)	CAR & ROC	C/S	Community	990	≥ 65	405	585	BMI	19.2		52.9	19.1/8.8	Multi
11.	Legesse M. et al., 2019 (34)	Ethiopia	C/S	Community	892	≥ 65	365	527	BMI	17.6		82.4		Multi
12.	Wondiye K et al., 2019 (35)	Ethiopia	C/S	Community	554	≥ 60	295	259	BMI	17.1		80.6	2.3	Not reported
13.	Agbozo, F. et al., 2018 (64)	Ghana	C/S	Community	120	60-70	42	78	BMI	10.0		51.7	21.7/16.6	Not reported
14.	Alao M. et al., 2015 (48)	Nigeria	Case-control	LTC	418	≥ 60	191	227	BMI	6.9		53.3	24.6/15.1	Not reported
15.	Tessfamichael D. et al., 2014 (40)	Ethiopia	C/S	Community	757	≥ 65	194	563	BMI	21.9		78.1		Multi
16.	Otitoola Oc. Et al., 2015 (57)	South Africa	Cohort	Elderly day care center	46	≥ 60	8	38	BMI	0		18.4	80.4	Not reported
17.	Pilleron, S. et al., 2015 (58)	CAR & ROC	C/S	Community	645	> 65	Not reported	Not reported	BMI	68.5		31.5		Not reported
18.	Esmayel EM. et al, 2013 (42)	Egypt	C/S	Community	200	≥65	112	88	NCL	56.0	18.0	26.0		Not reported
19.	Andre MB. et al., 2013 (59)	ROC	C/S	Community	370	≥60	194	176	MNA	13.8	28.4	57.8		Multi
20.	Adebusoye L. A. et al., 2012 (49)	Nigeria	C/S	Hospital	500	≥60	176	324	MNA	7.8	11.8	80.4		Multi
21.	Robb L. et al., 2017 (54)	South Africa	C/S	LTC	124	≥60	35	89	MNA	7.3	55.6	37.1		Not reported
22.	El-Sherbiny N.A. et al., al.2016 (41)	Egypt	C/S	Community	2,219	≥60	1165	1054	MNA-SF	10.9	41.9	47.2		Multi

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23.	Naidoo I. et al., 2015 (56)	South Africa	C/S	Community	984	≥60	224	760	MNA SF	51.0	5.5	43.4		Multi
24.	Adhana, Z K et al., 2015 (37)	Ethiopia	C/S	Community	423	≥60	99	324	MNA-SF	77.3	0	22.7		Multi
25.	Adebusoye L. et al., 2019 (50)	Nigeria	C/S	Hospital	350	>60	Not reported	Not reported	MNA-SF	4.57	95.43			Multi
26.	Marias ML. et al., 2007 (55)	South Africa	C/S	Community	210	≥60	53	157	MNA	5.7	47.1	47.1		Not reported
27.	Hailemariam H. et al., 2016 (36)	Ethiopia	C/S	Community	548	≥60	261	287	MNA	9.3	28.3	62.4		Not reported
28.	Abate et al., 2020 (38)	Ethiopia	C/S	Community	662	≥65	240	422	MNA	26.6	73.4			One
29.	Abdu AO et al., 2020 (39)	Ethiopia	C/S	Community	592	≥65	291	301	MNA	15.7	51.7	32.6		One
30.	Abd Allah et al., 2020 (43)	Egypt	C/S	Hospital	194	≥60	107	87	MNA	33.0	51.5	15.5		Multi
31.	Rasha El-Desouky 2017 (44)	Egypt	C/S	Community	320	≥60	167	153	MNA	35.0	38.4	26.6		Multi
32.	Khater MS et al., 2011 (45)	Egypt	C/S	LTC	120	≥60	56	64	MNA	10.8	40.8	48.4		Multi
33.	F. Mahjoub et al., 2019 (68)	Tunis	C/S	Hospital	40	≥65	21	19	MNA	40.0	52.5	7.5		One
34.	Adebusoye, LA. et al., 2018 (52)	Nigeria	C/S	Hospital	624	≥60	246	378	MNA	2.24	32.69	65.06		One
35.	Abd-El-Gawad et al., 2014 (51)	Nigeria	Cohort	Hospital	131	≥60	67	64	MNA	40.46	52.67	6.87		Multi
36.	Mahfouz, E.M. et al., 2013 (46)	Egypt	C/S	Community	350	≥60	136	214	MNA	8.6	29.7	61.7		Multi
37.	Andia A, et al., 2019 (70)	Niger	C/S	Community	384	≥60	132	252	MNA	7.8	58.5	33.7		Multi

Abbreviations: BMI = Body mass index
CAR = Central Africa Republic
ROC= Republic of Congo
C/S: Cross-section
MNA: Mini Nutritional Assessment Tool
MUAC: mid-upper arm circumference
LTC: Long term care
MNA-SF: Mini Nutritional Assessment Tool- short Form
NCL: Nutritional Screening Checklist

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Supplementary Table 3. Summary of the quality of included studies

S/no	Author & year of publication	Country	External validity quality Score				Internal validity quality Score						Total Score	Study Quality
			1	2	3	4	5	6	7	8	9	10		
1.	Aganiba, B. et al., 2015 (62)	Ghana	0	1	1	1	1	1	1	1	1	1	9	High
2.	Apprey C. et al., 2019 (63)	Ghana	0	1	1	1	1	1	1	1	1	1	9	High
3.	Diendéré, J. et al, 2018 (35)	Burkina Faso	0	1	1	1	1	1	1	1	1	1	9	High
4.	Mkhize X. et al., 2013 (53)	South Africa	0	1	1	1	1	1	1	1	1	1	9	High
5.	MJ, C. et al, 2012 (69)	Lake Victoria Basin of East Africa	0	1	1	1	1	1	1	1	1	1	9	High
6.	Ijarotimi OS & Keshinro OO, 2004 (65)	Tanzania	0	0	0	1	1	1	1	1	1	1	7	Moderate
7.	Nyaruhucha CNM. et al, 2001 (66)	Tanzania	0	1	1	1	1	1	1	1	1	1	9	High
8.	Olayiwola O.et al, 2006 (47)	Nigria	0	1	1	1	1	1	1	1	1	1	9	High
9.	Rouvray C. et al., 2014 (61)	Central Africa (COR & ROC)	0	1	1	1	1	1	1	1	1	1	9	High
10.	Legesse M. et al., 2019 (34)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
11.	Wondiye K et al., 2019 (35)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
12.	Agbozo, F. et al., 2018 (64)	Ghana	0	1	1	1	1	1	1	1	1	1	9	High
13.	Alao M. et al., 2015 (48)	Nigeria	0	0	0	1	1	1	1	1	1	1	7	Moderate
14.	Tessfamichael, D. et al., 2014 (40)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
15.	Otitoola Oc. Et al., 2015 (57)	South Africa	0	0	0	0	1	1	1	1	1	1	6	Moderate
16.	Pilleron, S. et al., 2015 (58)	Central Africa (COR & ROC)	1	1	1	1	1	1	1	1	1	1	10	High

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17.	Esmayel EM. et al, 2013 (42)	Egypt	0	1	1	1	1	1	1	1	1	1	9	High
18.	Andre MB. et al., 2013 (59)	Democratic Republic of Congo	0	1	1	1	1	1	1	1	1	1	9	High
19.	Adebusoye L.A. et al., 2012 (49)	Nigeria	0	1	1	1	1	1	1	1	1	1	9	High
20.	Robb L. et al., 2017 (54)	South Africa	0	1	1	1	1	1	1	1	1	1	9	High
21.	El-Sherbiny N.A. et al., al.2016 (41)	Egypt	0	1	1	1	1	1	1	1	1	1	9	High
22.	Naidoo I. et al., 2015 (56)	South Africa	0	1	1	1	1	1	1	1	1	1	9	High
23.	Adhana, Z K et al., 2015 (37)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
24.	Adebusoye L. et al., 2019 (50)		0	0	1	1	1	1	1	1	1	1	8	Moderate
25.	Marias ML. et al., 2007 (55)	South Africa	0	1	1	1	1	1	1	1	1	1	9	High
26.	Hailemariam H. et al., 2016 (36)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
27.	Abate et al., 2020 (38)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
28.	Abdu AO et al., 2020 (39)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
29.	Abd Allah et al., 2020 (43)	Egypt	0	0	0	0	1	1	1	1	1	1	6	Moderate
30.	Rasha El-Desouky_2017 (44)	Egypt	0	0	1	1	1	1	1	1	1	1	8	Moderate
31.	Khater MS et al., 2011 (45)	Egypt	0	0	0	0	1	1	1	1	1	1	6	Moderate
32.	F. Mahjoub et al., 2019 (68)	Tunis	0	0	0	1	1	1	1	1	1	1	7	Moderate
33.	Adebusoye, LA. et al., 2018 (52)	Nigeria	0	0	0	0	1	1	1	1	1	1	6	Moderate
34.	Abd-El-Gawad et al., 2014 (51)	Nigeria	0	0	0	1	1	1	1	1	1	1	7	Moderate

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35.	Mahfouz, E.M. et al., 2013 (46)	Egypt	0	0	0	1	1	1	1	1	1	1	7	Moderate
36.	Andia A, et al., 2019 (70)	Niger	0	0	0	0	1	1	1	1	1	1	6	Moderate

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	2
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	3
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	3-4
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	3
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	3
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	3
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	4
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	3-5
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	4
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	4
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	4
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	3
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	4
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	5
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	5
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	5
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	5
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	4
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	N/A

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	5
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	6
Study characteristics	17	Cite each included study and present its characteristics.	6
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	7
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	7-8
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	8
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	8-9
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	9
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	9
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	N/A
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	N/A
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	10
	23b	Discuss any limitations of the evidence included in the review.	12
	23c	Discuss any limitations of the review processes used.	12
	23d	Discuss implications of the results for practice, policy, and future research.	12
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	3
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	3
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	13
Competing interests	26	Declare any competing interests of review authors.	13
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	13

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Prevalence of malnutrition among old age people in Africa: Systematic Review and Meta-Analysis

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Abstract

Objective: To summarize available evidence and estimate the pooled prevalence of malnutrition among old age people in Africa.

Design: Systematic review and meta-analysis

Setting: Any health care or community research, reporting the prevalence or incidence of malnutrition in Africa from January 2000 to October 2021.

Participants: Old age people, aged above 60 years.

Outcome measures: Malnutrition either under or overnutrition.

Results: A total of 1,442 studies were retrieved based on the search strategy, where only 36 studies (n = 15,266 participants) reported from 11 African countries were included for meta-analysis. The reported prevalence of malnutrition ranges from 2.2 to 77.3% across the continent. Overall, the pooled prevalence of under and overnutrition were 18% (95% CI: 15-22; $I^2 = 98.1$; $p < 0.001$) and 33% (95% CI: 22-44; $p < 0.001$), respectively.

Conclusion: The prevalence of malnutrition in African old age people is high and differs by setting, assessment tool, and country of residence. Hence, due attention to geriatric nutrition is mandatory, and the need for a valid, reliable, and simple screening tool should be thought of.

Strengths and limitations of this study

- This provides a comprehensive estimate of the prevalence of malnutrition among old age

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people in Africa, including more recent studies conducted in the continent.

- Looked at the prevalence of malnutrition in the elderly using any of the validated screening tools.
- Only HINARI/PubMed and Google Scholars were retrieved.
- Limited to include only English literature.

Key words: Africa; Malnutrition; Old age people; Prevalence; Meta-analysis; Systematic-review

Introduction

Background

The global demographic structure of the population is changing dramatically, notably in Africa, due to several social and economic milestones [1]. These resulted in expanding population, improved life expectancy, and increased population size of older ages [1], [2]. The absolute number of old age people (60 years and above [3]) is expected to grow more rapidly in the coming decades than in any other part of the world [1].

The world is also in nutrition transition, a phenomenon where the changes in the foods and beverages consumed along with the reduced physical activity. In Africa, the risk of malnutrition among older people is a major challenge to the health care system and needs special and urgent attention. The usual food production practices, marketing style, and living standards have all contributed to an increase in the use of low-cost, packaged products that are high in fat, energy, and salt and yet low in nutritional quality. More importantly, poor infrastructure and limited resources, combined with conflict and poor access to healthcare services, are factors that contribute to the overwhelming levels of malnutrition and food insecurity on the continent [4]. The lack of policy direction for old age people, combined with prevalent chronic illnesses, adds a significant burden to the growing burden of malnutrition [5], [6]. Due to these and other aggravating factors, the government's commitment to ending all forms of malnutrition by 2030 [7] will be difficult to achieve.

While any age group may suffer from malnutrition, it is most common among old age people due to the changes in physiological, psychosocial, and health characteristics of individuals in this age group [8]. Furthermore, malnutrition has serious and life-threatening consequences that are known to be the main causes of increased morbidity and mortality among old age people [9]. Besides,

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complications of malnutrition, such as osteoarthritis, osteoporosis, diabetes, cardiovascular disease, and hypertension, inflict a significant social and economic burden on them [10].

Previous studies reported that the prevalence of malnutrition among old age people varies between 1.1% and 72.2% in different settings [11]–[15], where some of the variations could be attributed to differences in the measurement tools, study settings, and demographic groups that have been studied [16], [17]. Moreover, none of the previous reviews targeted the old age people in Africa [18]–[21], where the root causes of malnutrition are expected to be large. However, having concrete evidence on the burden of malnutrition using a more rigorous and systematic manner than individual pocket studies is the first step toward enhancing the elderly's interventions and policy direction. Context-specific evidence will provide better inform decision-makers in making informed decisions on addressing malnutrition among old age people for a better quality of life.

Objective

To estimate the prevalence of malnutrition among old age people in Africa according to the assessment tools, study setting, and regional location.

Methods

Search strategies

Before the beginning of this systematic review, we conducted a systematic search of review papers from the Cochrane Library, the International Prospective Register of Systematic Reviews (PROSPERO), and the Joanna Briggs Institute (JBI), and no review had been conducted on malnutrition among old age people in Africa. Yet, review protocol was not registered. Relevant publications were searched systematically from HINARI, PubMed, and Google Scholar sources from 01 January 2000 to 16 October 2021. We employed the search through relevant combinations of MeSH terms and related keywords, such as "malnutrition", "obesity", "overweight", "old age people", "older adults", "elderly", and "Africa" as indicated in **Supplementary File**. Lists of references from relevant studies were also manual-searched. Countries or territories included in Africa were defined according to the United Nations (UN) classification [22]. Studies were imported and checked for duplicates in Mendeley Desktop (Version 1.19.3).

Eligibility criteria

Inclusion criteria

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Based on the condition, context, and population (CoCoPop) framework [23] we included research publications in English language from January 1, 2000 to October 16, 2021 that reported the prevalence of malnutrition among old age persons aged 60 years and up in African countries. Intending to get a relatively up-to-date prevalence estimate and obtain a comparable age and malnutrition definition, we restricted studies conducted starting in 2000. All studies that reported the prevalence and/or incidence of malnutrition among old age people were considered regardless of the study design. When multiple studies reported the prevalence of malnutrition based on the same study, the one with the larger sample size was included.

Exclusion criteria

Qualitative studies, study protocols, abstracts, opinions, commentaries, case reports, review papers, and studies on African-Americans who do not live on the African continent were all excluded from the review. Publications reporting malnutrition in post-intervention were excluded. Duplicated articles and articles without access to the full text were removed from the review.

Outcome measuring tools

Studies that measured malnutrition with any standard and validated tools, including questionnaires such as Patient-Generated Subjective Global Assessment (PG-SGA), Mini Nutritional Assessment (MNA), Subjective Global Assessment tools (SGA), or Nutritional Screening Checklist (NCL), anthropometric [mid-upper arm circumference (MUAC), calf-circumference (CC), body mass index (BMI), etc.], or biochemical indices (serum haemoglobin, albumin, etc.) were considered. We used the reported prevalence of malnutrition or undernutrition based on the standard definition set for each nutritional assessment method. When the entire text of an article could not be obtained or if there was missing information, the corresponding author(s) was contacted through email. Requested full-text papers that were not submitted until October 31, 2021, and the study was discarded.

Study selection and quality assessment

A two-step process was used to select the relevant publications. First, the scan of the titles and abstracts was conducted for the inclusion of references that meet the eligibility criteria. The entire text was retrieved when any doubts existed about the relevance of a study. Second, the full texts were then used to make a judgement on whether or not the study should be included in the review.

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The two authors work individually on all components of the selection process. Disagreements were settled through conversation.

Selected studies were evaluated using a ten-item rating checklist designed for research reporting prevalence [24], where better quality articles have a higher score as indicated in **Supplementary Table 1**.

Data extraction

The two authors separately extracted data from the full-text articles using a pre-defined standard format produced in Excel. The extract comprised the following information: general (authors, year of publication, and country), participants (age, gender, and morbidity status), and study details (study design, study setting, sample size, measurement tool used, and outcome measures). Morbidity was coded as one or multi when a single or multiple underlying health conditions were an inclusion criterion.

Statistical analysis

The pre-coded extracted data in excel format was exported to STATA software version 14 [25] for descriptive and meta-analysis. For each study, the unadjusted prevalence of malnutrition and standard errors were calculated, and a logarithmic (log) transformation for proportional data was used to reduce variance and obtain the pooled prevalence as an effect size (ES). Considering the level of methodological heterogeneity and variance within the study, a random effect model of the DerSimonian and Laird [26] method was used to conduct the meta-analysis.

The heterogeneity between studies was assessed using a forest plot, Cochran's Q test, and I-square (I^2) test. The I^2 values of 25, 50, and 75% with a *p-value* less than 0.10 represent low, medium, and high heterogeneity, respectively [27]. Publication bias was checked graphically in a funnel plot supplemented by Egger's statistics [28]. We also applied a subgroup analysis, a sensitivity analysis, and meta-regression to identify the potential sources of heterogeneity. A statistically significant effect size estimate has a P-value of less than 0.05.

Patient and public involvement

No patient and public involvement.

Results

Search results

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A total of 1,442 articles were obtained through searches, of which 206 were removed due to duplicates. With the title and abstract screening, 971 articles were removed depending on the focus of the review (older age in the African continent), relevant nutritional assessment method, and language (other than English). Furthermore, 237 articles were retrieved during the full-text screening. Finally, 37 and 36 articles were selected for qualitative (systematic review) and quantitative analysis (meta-analysis), respectively. The comprehensive selection process presented using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement [29] (**Figure 1).**

Characteristics of included studies

A total of 37 studies reported the prevalence of malnutrition among old age people. While 34 studies (92%) were published in the last decades between 2011 and 2020, the remaining 8% were issued between 2001 and 2010. All the included studies were reported from 11 African countries, where more than half (65%, n = 24) were from four countries: Ethiopia, Egypt, Nigeria, and South Africa.

Seven studies (18.9%) were from Ethiopia [30]–[36] and six studies (16.2%) each from Egypt [37]–[42] and Nigeria [43]–[48]. While five studies (13.5%) were from South Africa [49]–[53], four studies (10.8%) were from the Central African Republic (CAR) and the Republic of Congo [54]–[57] three studies (8.1%) from Ghana [58]–[60], and two studies (6.06%) were from Tanzania [61], [62].

Except for three studies (two cohorts [31], [47] and one case-control [44]), all other identified studies were cross-sectional study designs. Malnutrition prevalence was reported in four studies as a comparison between institutional versus noninstitutional [44], [61], high versus low economic classes [50], and with and without dementia [57]. However, one study from Tanzania examined the nutritional status after a post-supplemental food program [63] and was excluded from the meta-analysis. Majority of the studies (n = 22) were conducted in community settings, while 11 studies were conducted among institutionalized elders.

Moreover, a total of 15,266 study participants were found, of whom more than half (58%) were females. The selected studies had sample sizes ranging from 40 [64] to 2,219 [37] people. Regarding the nutritional assessment tools used, more than half of the studies reported malnutrition based on either the full-or short-form of the Mini Nutritional Assessment tool (MNA-FF: n = 15,

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MNA-SF: n = 4). While 18 studies employed various anthropometric dimensions such as BMI, MUAC, and CC separately or in combinations to assess malnutrition. One study used an interviewer- administered Nutritional Screening Checklist (NCL)[38] and another study utilized a combination of MNA and the geriatric nutrition risk index (GNRI)[64]. Two-thirds (n = 23) of the studies reported single (n = 8) or multiple (n = 15) underlying clinical conditions such as fracture, heart disease, dementia, Parkinson's disease, or cancer (**Supplementary Table 2**).

Quality of the studies

According to the quality assessment checklist, studies were either in moderate (6-8 points) [39]–[42], [44], [46]–[48], [53], [61], [64], [65],or high-quality groups (9 or 10 points) [30]–[38], [43], [45], [49]–[52], [54], [55], [57]–[60], [62], [66]. Study bias was mostly exacerbated as a result of convenience sampling and a low response rate among study participants. In almost all studies, the possibility of external validity was low as the target population was not nationally representative of the country. However, the quality of two studies [56], [63] was not assessed and hence not processed for quantitative data analysis. The first was [56] due to duplication of results from previous data [57] and the latter [63] was due to the outcome variable being measured after an intervention (**Supplementary Table 3**).

Prevalence of malnutrition

The prevalence of malnutrition varied highly across the studies conducted in different countries, assessment tools used, and healthcare settings. We presented an overview of the findings from various studies as under and overnutrition using two assessment tools in the section below.

Prevalence of undernutrition

Seventeen studies from hospitals (n = 5), the communities (n = 10), a long-term care center (n = 1), and a daycare center (n = 1), utilized body mass index (BMI) as a nutritional assessment tool. As low as zero [53] and 4.0% [49] of research participants respectively from the elderly daycare center and hospital settings in South Africa were classified as undernourished (BMI $\leq$ 18.5 kg/m^2). Furthermore, the highest prevalence of undernutrition (25.6%) was reported in a Tanzanian hospital [62]. From the community settings, undernutrition was reported at 9.9% in Ghana [59] and 26.4% in the East Africa [66].

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On the other hand, 20 studies from hospitals (n = 7), communities (n = 11), and long-term care centers (n = 2), utilized mini nutritional assessment (MNA) as a nutritional assessment tool. The outcome was reported as at risk of undernutrition, undernutrition, or normal status. The results showed that old age people in the community were more victims of undernutrition compared to hospitals and long-term care centers. The prevalence ranged from 5.7% in Niger [65] to 56% in Egypt [38] and 18% in Egypt [38] to 58.5% in Niger [65], respectively. While in Nigerian hospitals, 2.24% [48] to 40.46% [47] and 11.8% [45] to 52.7% [47] of old age people were undernourished or at risk of malnutrition, respectively. Besides, 7.3% in South Africa [50] and 10.8% in Egypt [41] were undernourished using MNA from the long-term care center. Similarly, 10.9% of community study participants in Egypt [37] and 77.3% in Ethiopia [33] were undernourished, while none from Ethiopia[33] and 41.9% from Egypt [37] were at risk of undernutrition using MNA-SF. Moreover, 4.57% of hospital study participants in Nigeria [46] were undernourished, while 95.43% were either normal or at risk of undernutrition.

Prevalence of overnutrition

Overnutrition was reported in 17 studies that used BMI as a nutritional assessment tool. Accordingly, 77% of the elderly people studied in a South African hospital [49] were overweight or obese. There were none from Tanzania [62] in a comparable situation. While 15.3% of study participants in East Africa [66] and 42.1% in Ghana [59] were affected by overweight or obesity in the community settings.

Meta-analysis

Adequate data were available to conduct pooled malnutrition prevalence estimates for two nutrition screening tools: under nutrition from 36 studies (BMI = 16 and MNA = 20), and overnutrition from 16 studies using only BMI. Since the full-and short-form of MNA and NCL tools are similar in the interpretation of the outcome variable, they were treated as similar tools. The random pooled prevalence of under and overnutrition in Africa were 18% (95% CI: 15-22; p < 0. 001) and 33% (95% CI: 22-44; p < 0. 001), respectively **(Figure 2 and 3)**.

A high level of statistical heterogeneity was observed between studies (heterogeneity of Cochran Q statistics = 1828.98 with *df* = 35 and *p-value* < 0.001; I^2 = 98.1%; *Tau-square* (τ^2 = 0.01).

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Moreover, a minimal publication bias has been noted in the published studies. Studies with lower standard error and larger sample size might be included, as indicated in of the funnel and Egger's publication plots (**Figure 4**). Smaller studies with larger standard errors plotted on the right of the funnel plot tend to have both smaller and larger odds ratios. This may indicate the existence of bias due to small-study effects. Moreover, the Egger plot shows that the data near the origin are unsystematically elevated, where the confidence interval (CI) does not include zero, indicating asymmetry in the funnel plot and evidence of publication bias.

Furthermore, Egger's regression asymmetry test value was 3.50 (p = 0.008), suggesting publication bias. Here, the positive coefficient indicates that small studies overestimate the effect size. However, no study appeared to have a significant impact on the summary effect size in the sensitivity analysis, strongly suggesting that the effect estimate is consistent across groups and potentially giving a reliable estimate of the outcome. In addition, the Trim and Fill analysis was performed to examine the potential impact of publication bias on the interpretation of the data. The method adds 12 "missing" studies to the dataset. However, the correction for publication bias does not change the overall interpretation of the findings.

Subgroup analysis and meta-regression

Subgroup analysis and meta-regression were performed for the study regions, settings, assessment tools, age group, and presence of multimorbidity using the random-effects inverse-variance model with the DerSimonian-Laird estimate. In general, a relatively high prevalence of malnutrition was observed in central Africa (3.76%; 95% CI: 3.22-4.39) as compared to other regions of the continent. Similarly, the community elderly (3.12%; 95% CI: 2.66-3.67), older age groups (3.54%; 95% CI: 2.32–5.41), and those with no reported clinical condition (3.16%; 95CI: 2.62–3.80) had a higher malnutrition burden than their counterparts. The prevalence estimates had a statistically significant difference among all subgroups except for the assessment tools used (p = 0.430) (Table 1).

Table 1: Subgroup analysis for the prevalence of malnutrition among old age people, 2021

Subgroup	No of studies included	Prevalence with 95%CI	Cochran's Q statistics for heterogeneity	P value	I^2
Regions					

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Northern Africa	8	2.82[2.72-2.92]	3.93	0.788	0.0%
Western Africa	10	2.46[1.95-3.11]	263.06	< 0.001	96.6%
Central Africa	3	3.76[3.22-4.39]	2.08	0.354	3.7%
Eastern Africa	10	3.46[2.29-5.22]	312.49	< 0.001	97.1%
Southern Africa	5	1.79[1.41-2.26]	60.54	< 0.001	93.4%
Between groups			42.08	< 0.001	
Study settings					
Community	21	3.12[2.66-3.67]	829.56	< 0.001	97.6%
Hospital	11	2.12[1.67-2.70]	132.88	< 0.001	92.5%
Long-Term Care	3	2.34[2.08-2.63]	0.31	0.859	0.0%
Elderly day care center	1	1.04[0.87-1.24]	0.00	-	-
Between groups			86.79	< 0.001	
Assessment Tools					
BMI	16	2.98[2.37-3.75]	213.68	< 0.001	93.0%
NCL	1	5.74[0.71-46.74]	-	-	-
MNA	15	2.45[2.11-2.85]	264.09	< 0.001	94.7%
MNA-SF	4	2.45[1.96-3.06]	136.91	< 0.001	97.8%
Between groups			2.76	0.430	
Age group					
≥ 60	26	2.54[2.17-2.98]	1102.79	< 0.001	97.7%
≥ 65	10	3.54[2.32-5.41]	361.43	< 0.001	97.5%
Between groups			2.06	< 0.001	
Morbidity					
One	7	1.66[1.46-1.89]	91.58	< 0.001	93.4%
Multi	16	2.84[2.50-3.22]	205.87	< 0.001	92.7%
Not reported	13	3.16[2.62-3.80]	36.06	< 0.001	66.7%
Between groups			16.44	< 0.001	
Overall	**37**	**2.77(2.43-3.16)**	**1573.48**	**< 0.001**	**97.7%**

However, only the study setting was statistically significant (adjusted R-squared = 39.2%, p = 0.049) predicting the prevalence of malnutrition in the meta-regression analysis.

Discussion

We conducted this review paper with the intention of estimating the pooled prevalence of malnutrition disaggregated by relevant factors such as study setting, screening tool, and so on. According to the findings, the prevalence of malnutrition ranges from 4.0% [49] to 77.3% [33] across countries, depending on the assessment tools used and the study settings. These prevalence figures are consistent with previous European estimates of 3.8 to 67.4% [19], [21] among older people despite differences in the settings. In contrast, the highest risk of malnutrition was observed in the community-dwelling old age people assessed by the MNA (both full and short forms) than in other studies; 7.5% to 77.3% versus 5% [18] to 19% [67]. The variation may be due to the economic variables of the countries.

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Overall undernutrition was 17.1% in the African continent using BMI, which is comparable to European countries, 3.8-18.2% [21]. While 27% of the study participants in the continent were overweight or obese. Furthermore, many studies (n=14) in our review used BMI as a nutritional assessment tool for old age people. Similar studies in Africa still preferred to use BMI for old age people where 19.9% and 30% of subjects were undernourished, overweight or obese, respectively [68].

However, the BMI cutoff point used to identify underweight can have a significant impact on prevalence estimates, which tend to misclassify the elderly with physical spinal deformities and other factors that can affect height measurements [69]. Due to such measurement errors encountered, the assessment of nutritional status in the old age people become a challenging task, which warrants more comprehensive and multidimensional assessments including clinical examination, anthropometric measures, laboratory tests, dietary surveys, and social aspects [70], [71]. This means one assessment tool may not be suitable for all settings, which need contextualization [18]. Though more than 22 best-validated nutrition screening tools were identified for older adults [17], [19], there has long been a lack of consensus regarding the criteria needed to make a diagnosis of malnutrition.

On the other hand, the meta-analysis revealed a pooled prevalence of malnutrition of 18% (95% CI: 15-22; $p < 0.001$). This is in the range of the ones reported from the world, ranging from 0.8 to 24.6% [72] depending on the study regions, settings, and assessment tools used. The increasing number of an ageing population in Africa and nutritional transitions might have contributed to a higher burden of malnutrition. Our estimate is also consistent with the pooled prevalence of Indian studies (18.29%; 95% CI: 15.24-21.57) [73]. However, the result is slightly higher than the prevalence of malnutrition reported from the Central Demographic Republic of Congo and Nigeria (14.5%; 95% CI: 0.0-40.4%) [72]. This disagreement could be explained by the use of various assessment procedures and the merging of pooled estimates from two countries where the burden of malnutrition might be higher.

Surprisingly, the current prevalence estimate is lower than the reviews reported from European older persons (48.4%; 95% CI: 41.5-51.8) [19]. The disparity could be attributed to economic factors, health literacy, and healthcare access, and the majority of the previous review data came from hospital settings, implying that individuals with higher health risks were included.

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Still, there was significantly high heterogeneity between the studies (Q = 1873.99; df = 36; p< 0.001; T^2= 0.01; I^2 = 98.1%). Sensitivity analysis did not improve this heterogeneity, which is consistent with other previous findings [18], [72]. Because of the highly complex physiological, social, and temporal nature of malnutrition, which differs from person to person, as well as bias introduced by research design, analysis of the pooled prevalence of malnutrition is difficult to account for all sources of variation [74].

In case of subgroup analysis, the frequency of malnutrition showed a great variation depending on the study regions, study settings, age groups, and clinical conditions of study participants. Malnutrition was found to be prevalent in central Africa (3.76%), in older age groups (3.54%), with no documented clinical condition (3.16%), and in community-dwellers (3.12%). The prevalence was statistically significant in all subgroups except between the assessment tools used (p = 0.430). These findings are consistent with the reviews from India [73] and the rest of the world [72]. While the meta-regression identified the study setting as a statistically significant (adjusted R^2 = 39.20%, p = 0.049) predictor of malnutrition prevalence. This is, however, different from the reviews conducted in India [73] and the world [72] where the study region (R^2= 27.1%, p = 0.026) and assessment tools (R^2= 76.47%, p<0.001) were statistically significant predictors of the malnutrition prevalence, respectively. The variation might be due to the difference in assessment tools used between the studies.

Strength and Limitation

To our knowledge, this is the first systemic review and meta-analysis that provides a comprehensive estimate of the prevalence of malnutrition among old age people in Africa, including more recent studies conducted in the continent. Even though this review has its strengths, it is not without limitations. Due to the inherent heterogeneity among the included studies (I^2 = 98.08%), the interpretation of the current study needs precaution. Being limited to including only English literature and not including research published in books, magazines, and others might limit the generalizability of the estimate.

Conclusion

This systematic review and meta-analysis showed high malnutrition prevalence among old age people which warrants a targeted policy intervention. The prevalence varied greatly depending on

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the study setting, the tool used, and the country of residence. Due attention to geriatric nutrition is mandatory, and to better monitor and evaluate, the use of a more reliable, valid, and standardized malnutrition assessment tool is strongly recommended.

List of abbreviations

BMI: Body Mass Index; CC: Calf Circumference; HINARI: Health InterNetwork Access to Research Initiative; MeSH: Medical Subject Headings; MNA: Mini Nutritional Assessment

Declarations

Ethical approval and formal consent

Not applicable

Consent for publication

Not applicable

Availability of data and materials

All data generated or analyzed during this study are included in this published article and its supplementary information files, tables, and figures.

Competing interests

The authors declare that they have no competing interests

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Author Contributions

The study was created and designed by AM under the supervision and approval of NF. They are both responsible for all parts of this work. The final manuscript was read and approved by both writers.

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Figure legends

Figure 1: PRISMA flow-chart of studies screened and reviewed.

Figure 2: Forest plot displaying the pooled prevalence of undernutrition among old age people in Africa.

Figure 3: Forest plot displaying the pooled prevalence of overnutrition among old age people in Africa.

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Figure 4: Probability of publication bias for included studies (a) funnel plot and (b) Egger's publication bias plot.

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Figure Legends

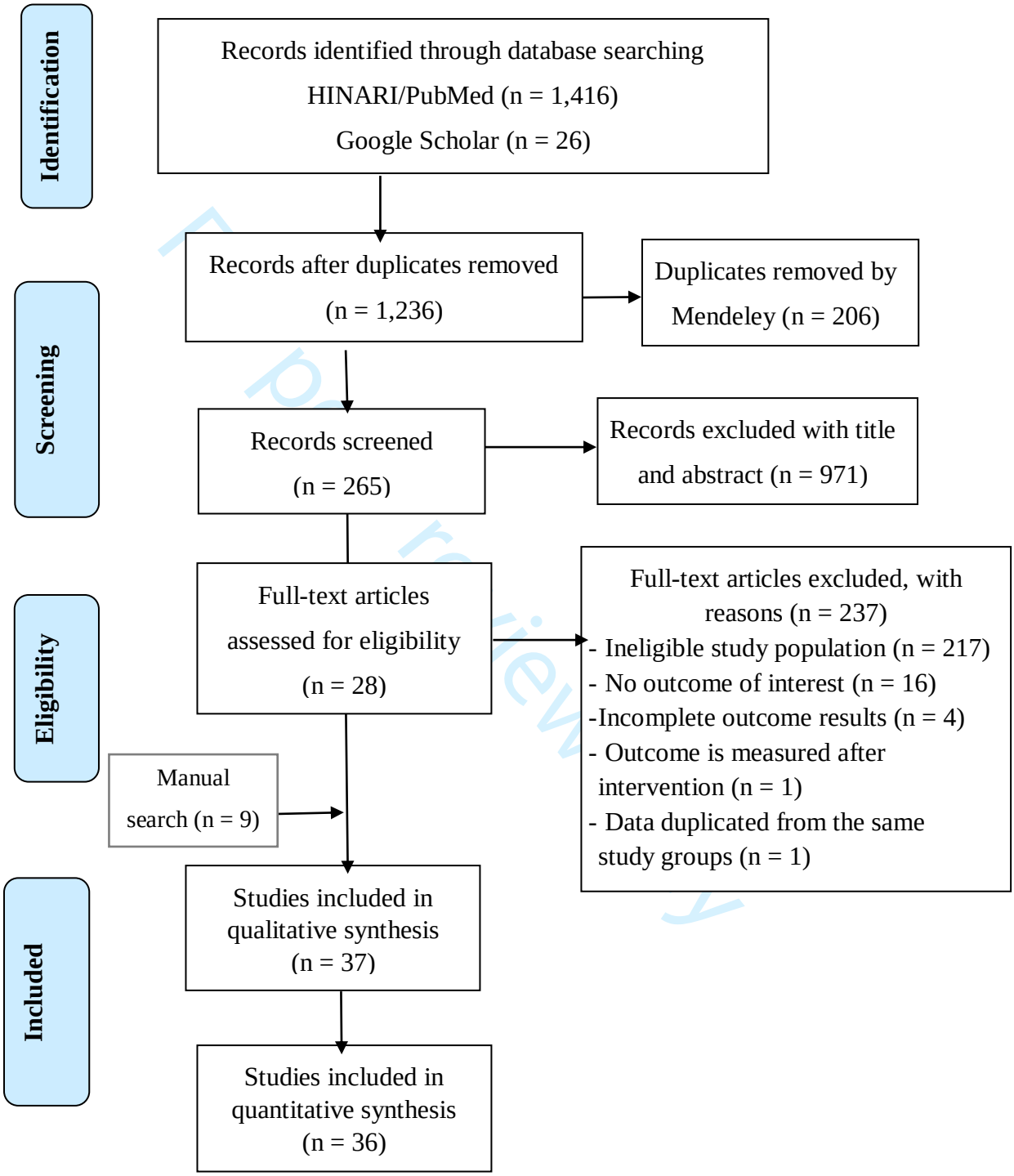


Figure 1: PRISMA flow-chart of studies screened and reviewed.

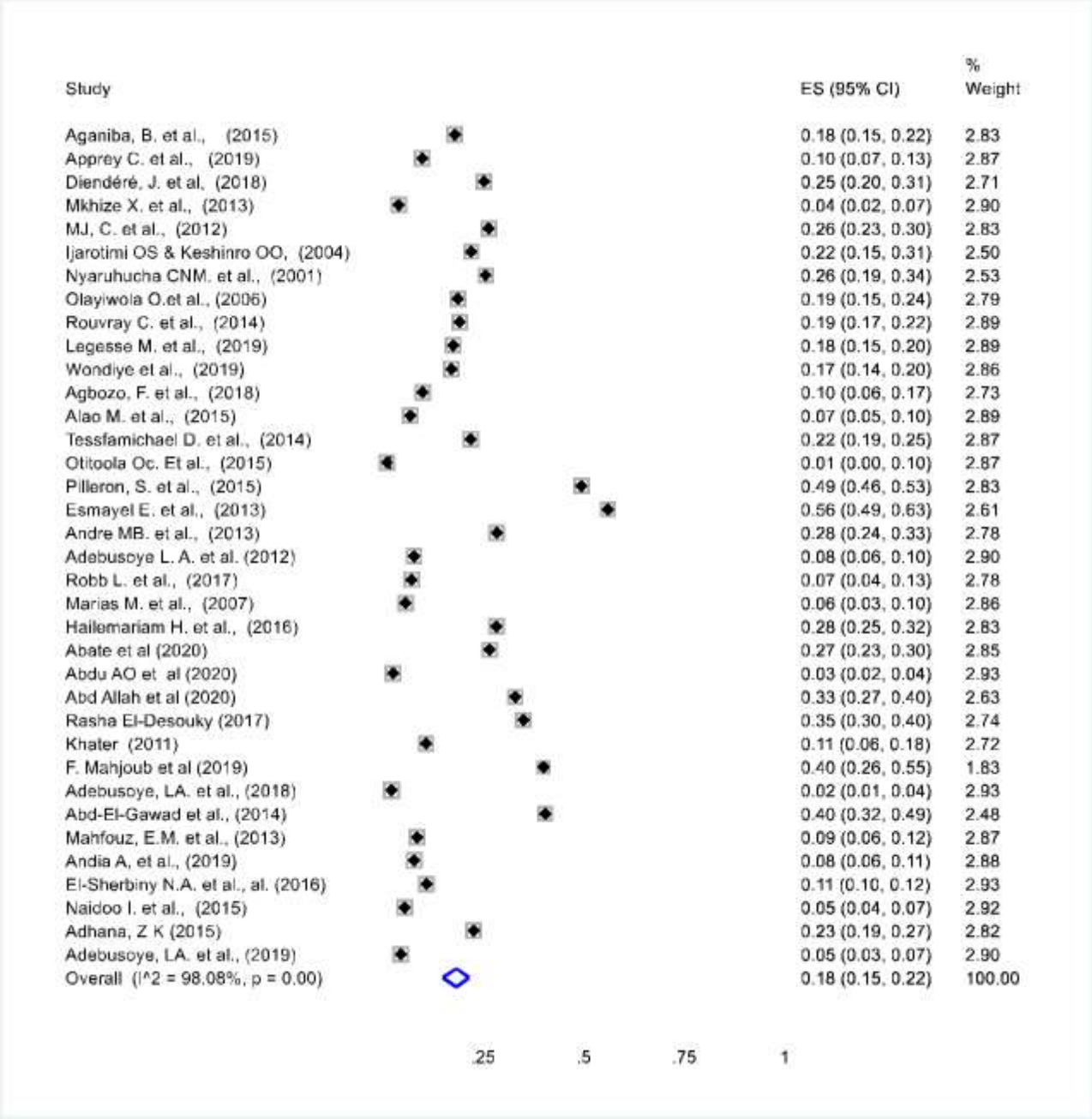


Figure 2: Forest plot displaying the pooled prevalence of undernutrition among old age people in Africa.

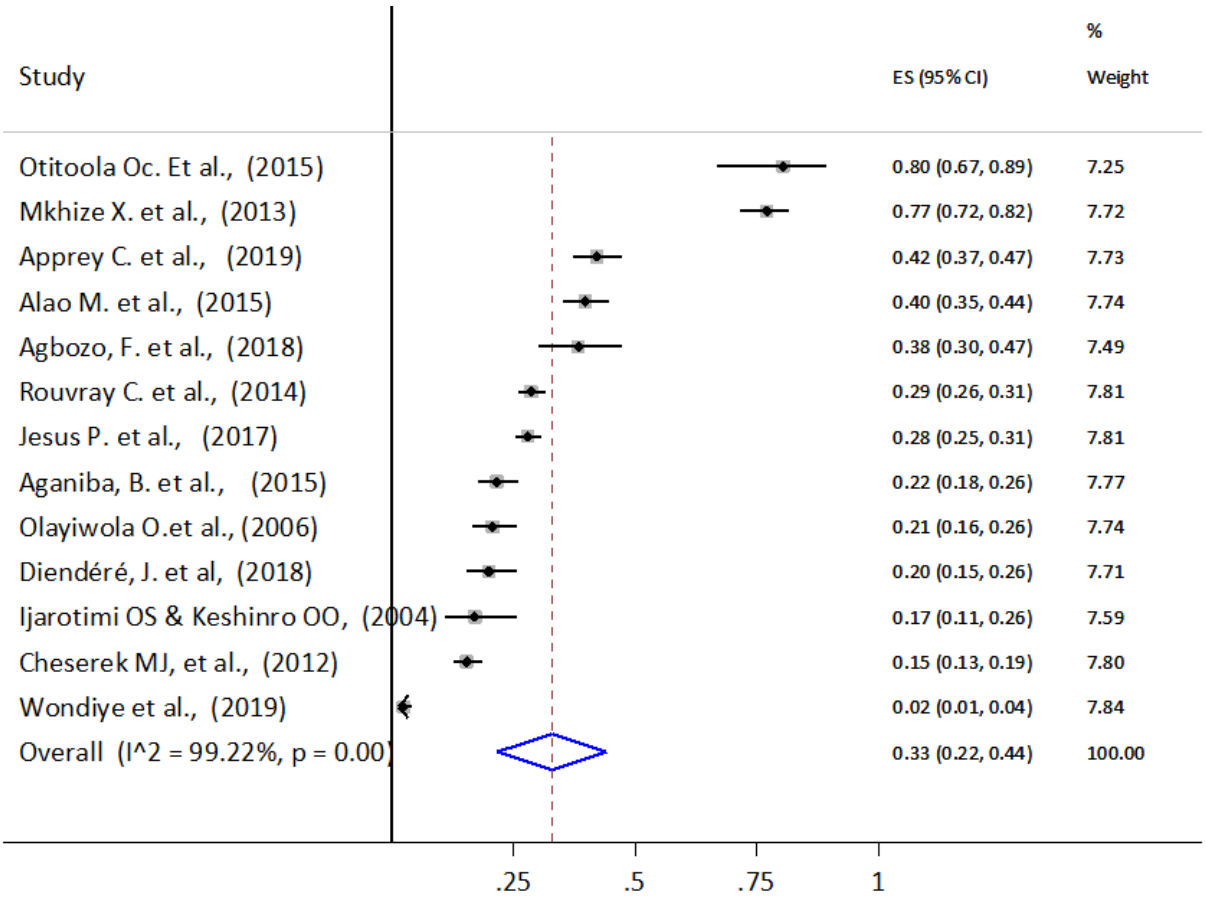
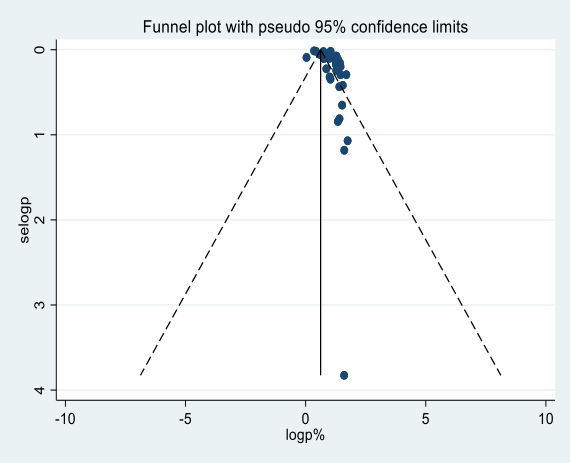


Figure 3: Forest plot displaying the pooled prevalence of overnutrition among old age people in Africa.

iew only

a.



b.

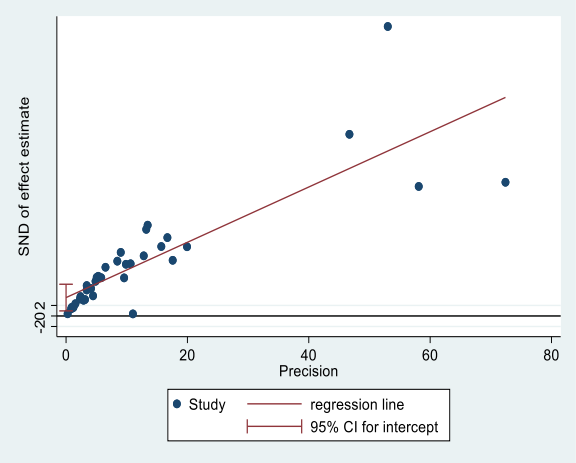


Figure *4:* Probability of publication bias for included studies (a) funnel plot and (b) Egger's publication bias plot.

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Supplementary File

Search items for HINARI/PubMed

#1= "Prevalence"[MeSH] OR "Epidemiology"[MeSH]

#2= "Malnutrition"[MeSH] OR "Obesity"[MeSH]

#3= malnutrition [tiab] OR nutrition*[tiab] OR obesity [tiab]

#4= aged [MeSH] OR elder [MeSH]

#5= old*[tiab] OR aged [tiab] OR elder [tiab] OR geriatric [tiab]

#6= Africa [MeSH] OR Africa [tiab]

#7= Angola[tiab] OR Benin[tiab] OR Botswana[tiab] OR Burkina Faso[tiab] OR Burundi[tiab] OR Cameroon[tiab] OR Cape Verde[tiab] OR Central African Republic[tiab] OR Chad[tiab] OR Comoros[tiab] OR Congo[tiab] OR Côte d'Ivoire[tiab] OR Democratic Republic of the Congo[tiab] OR Djibouti[tiab] OR Equatorial Guinea[tiab] OR Eritrea[tiab] OR Ethiopia[tiab] OR Gabon[tiab] OR Gambia[tiab] OR Ghana[tiab] OR Guinea[tiab] OR Guinea-Bissau[tiab] OR Kenya[tiab] OR Lesotho[tiab] OR Liberia[tiab] OR Madagascar[tiab] OR Malawi[tiab] OR Mali[tiab] OR Mauritania[tiab] OR Mauritius[tiab] OR Mayotte[tiab] OR Mozambique[tiab] OR Namibia[tiab] OR Niger[tiab] OR Nigeria[tiab] OR Réunion[tiab] OR Rwanda[tiab] OR Saint Helena[tiab] OR São Tomé and Príncipe[tiab] OR Senegal[tiab] OR Seychelles[tiab] OR Sierra Leone[tiab] OR Somalia[tiab] OR South Africa[tiab] OR Sudan[tiab] OR Swaziland[tiab] OR Togo[tiab] OR Uganda[tiab] OR United Republic of Tanzania[tiab] OR Zambia[tiab] OR Zimbabwe[tiab] OR Algeria[tiab] OR Egypt[tiab] OR Libya[tiab] OR Morocco[tiab] OR Tunisia[tiab]

#8=#2 OR #3

#9= #4 OR #5

#10= #6 OR #7

#11= #1 AND #8 AND #9 AND #10

#12= Filter: date from 2000/01/01 to 2019/12/10; English language; Humans

Combined search terms

((("Prevalence"[MeSH] OR "Epidemiology"[MeSH]) AND ("Malnutrition"[MeSH] OR "Obesity"[MeSH] OR malnutrition [tiab] OR nutrition*[tiab] OR obesity[tiab]) AND (aged[MeSH] OR elder[MeSH] OR old*[tiab] OR aged[tiab] OR elder[tiab] OR geriatric[tiab]) AND (Africa[MeSH] OR Africa[tiab] OR Angola[tiab] OR Benin[tiab] OR Botswana[tiab] OR Burkina Faso[tiab] OR Burundi[tiab] OR Cameroon[tiab] OR Cape Verde[tiab] OR Central

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African Republic[tiab] OR Chad[tiab] OR Comoros[tiab] OR Congo[tiab] OR Côte d'Ivoire[tiab] OR Democratic Republic of the Congo[tiab] OR Djibouti[tiab] OR Equatorial Guinea[tiab] OR Eritrea[tiab] OR Ethiopia[tiab] OR Gabon[tiab] OR Gambia[tiab] OR Ghana[tiab] OR Guinea[tiab] OR Guinea-Bissau[tiab] OR Kenya[tiab] OR Lesotho[tiab] OR Liberia[tiab] OR Madagascar[tiab] OR Malawi[tiab] OR Mali[tiab] OR Mauritania[tiab] OR Mauritius[tiab] OR Mayotte[tiab] OR Mozambique[tiab] OR Namibia[tiab] OR Niger[tiab] OR Nigeria[tiab] OR Réunion [tiab] OR Rwanda[tiab] OR Saint Helena[tiab] OR São Tomé and Príncipe[tiab] OR Senegal[tiab] OR Seychelles[tiab] OR Sierra Leone[tiab] OR Somalia[tiab] OR South Africa[tiab] OR Sudan[tiab] OR Swaziland[tiab] OR Togo[tiab] OR Uganda[tiab] OR United Republic of Tanzania[tiab] OR Zambia[tiab] OR Zimbabwe[tiab] OR Algeria[tiab] OR Egypt[tiab] OR Libya[tiab] OR Morocco[tiab] OR Tunisia[tiab])))

In case of Google Scholar advanced search for the topic was as follows:

1. With all of the words: old age people, or elderly
2. With at least one of the words: prevalence, epidemiology, malnutrition, undernutrition, overnutrition, obesity, overweight, old age people, older adult, elderly, Africa, Sub-Saharan Africa, Horn of Africa
3. Where my words occur: in the title of the article

Supplementary Table 1: Quality appraisal checklist for prevalence studies

Items		**Quality score**
	External validity Criteria	
1.	Was the study's target population a close representation of the national population in relation to relevant variables?	(1 point)
2.	Was the sampling frame a true or close representation of the target population?	(1 point)
3.	Was some form of random selection used to select the sample, OR was census undertaken?	(1 point)
4.	Was the likelihood of nonresponse bias minimal?	(1 point)
		Total (4 points)
	Internal validity Criteria	
5.	Were data collected directly from the subjects (as opposed to a proxy)?	(1 point)
6.	Was an acceptable case definition used in the study?	(1 point)
7.	Was the study instrument that measured the parameter of interest shown to have validity and reliability?	(1 point)
8.	Was the same mode of data collection used for all subjects?	(1 point)

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9.	Was the length of the shortest prevalence period for the parameter of interest appropriate?	(1 point)
10.	Were the numerator(s) and denominator(s) for the parameter of interest appropriate?	(1 point)
		Total (6 points)
	Overall total	10 points

Methodological quality

High quality: score > 8

Moderate quality: score 6-8

Low quality: score 0-5

(Source: Hoy et al.(24))

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Supplementary Table 2. Characteristics of Selected Studies Using Different Nutritional Assessment Tools

S/no	Authors	Country	Study design	Study setting	Sample size	Age group	Gender		Assessment tool used	Nutritional status (outcome variable) in %				Multi-morbidity
							Male	Female		Under-nutrition	At risk of malnutrition	Normal-weight/ Nutrition	Overweight/ obesity	
1.	Aganiba, B. et al., 2015 (62)	Ghana	C/S	Community	400	≥ 65	177	223	BMI	18.0		60.5	21.5	Not reported
2.	Apprey C. et al., 2019 (63)	Ghana	C/S	Community	375	≥ 60	127	248	BMI	9.9		48	30.9/11.2	Multi
3.	Diendéré, J. et al, 2018 (35)	Burkina Faso	Cohort	Hospital	222	≥ 60	121	101	BMI	25.2		55.0	14.9/5.0	One
4.	Mkhize X. et al., 2013 (53)	South Africa	C/S	Hospital	270	≥ 60	46	224	BMI	4.0		19.0	24.0/53.0	One
5.	MJ, C. et al, 2012 (69)	Lake Victoria Basin of East Africa	C/S	Community	537	≥ 60	227	310	BMI	26.4		58.3	10.8/4.5	Not reported
6.	Ijarotimi & Keshinro, 2004 (65)	Tanzania	C/S	Hospital	100	≥ 60	51	49	BMI	22.0		61.0	17.0	Not reported
7.	Nyaruhucha CNM. et al, 2001 (66)	Tanzania	C/S	Hospital	121	≥ 60	67	54	BMI	25.6		74.4	0	Not reported
8.	Olayiwola O.et al, 2006 (47)	Nigeria	C/S	Hospital	295	≥ 60	153	142	BMI	18.7		60.7	15.7/4.9	Not reported
9.	Jésus p. et al., 2017 (60)	CAR & ROC	C/S	Community	990	≥ 65	405	585	BMI	19.2		52.9	19.1/8.8	Multi
10.	Rouvray C. et al., 2014 (61)	CAR & ROC	C/S	Community	990	≥ 65	405	585	BMI	19.2		52.9	19.1/8.8	Multi
11.	Legesse M. et al., 2019 (34)	Ethiopia	C/S	Community	892	≥ 65	365	527	BMI	17.6		82.4		Multi
12.	Wondiye K et al., 2019 (35)	Ethiopia	C/S	Community	554	≥ 60	295	259	BMI	17.1		80.6	2.3	Not reported
13.	Agbozo, F. et al., 2018 (64)	Ghana	C/S	Community	120	60-70	42	78	BMI	10.0		51.7	21.7/16.6	Not reported
14.	Alao M. et al., 2015 (48)	Nigeria	Case-control	LTC	418	≥ 60	191	227	BMI	6.9		53.3	24.6/15.1	Not reported
15.	Tessfamichael D. et al., 2014 (40)	Ethiopia	C/S	Community	757	≥ 65	194	563	BMI	21.9		78.1		Multi
16.	Otitoola Oc. Et al., 2015 (57)	South Africa	Cohort	Elderly day care center	46	≥ 60	8	38	BMI	0		18.4	80.4	Not reported
17.	Pilleron, S. et al., 2015 (58)	CAR & ROC	C/S	Community	645	> 65	Not reported	Not reported	BMI	68.5		31.5		Not reported
18.	Esmayel EM. et al, 2013 (42)	Egypt	C/S	Community	200	≥65	112	88	NCL	56.0	18.0	26.0		Not reported
19.	Andre MB. et al., 2013 (59)	ROC	C/S	Community	370	≥60	194	176	MNA	13.8	28.4	57.8		Multi
20.	Adebusoye L. A. et al., 2012 (49)	Nigeria	C/S	Hospital	500	≥60	176	324	MNA	7.8	11.8	80.4		Multi
21.	Robb L. et al., 2017 (54)	South Africa	C/S	LTC	124	≥60	35	89	MNA	7.3	55.6	37.1		Not reported
22.	El-Sherbiny N.A. et al., al.2016 (41)	Egypt	C/S	Community	2,219	≥60	1165	1054	MNA-SF	10.9	41.9	47.2		Multi

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23.	Naidoo I. et al., 2015 (56)	South Africa	C/S	Community	984	≥60	224	760	MNA SF	51.0	5.5	43.4		Multi
24.	Adhana, Z K et al., 2015 (37)	Ethiopia	C/S	Community	423	≥60	99	324	MNA-SF	77.3	0	22.7		Multi
25.	Adebusoye L. et al., 2019 (50)	Nigeria	C/S	Hospital	350	>60	Not reported	Not reported	MNA-SF	4.57	95.43			Multi
26.	Marias ML. et al., 2007 (55)	South Africa	C/S	Community	210	≥60	53	157	MNA	5.7	47.1	47.1		Not reported
27.	Hailemariam H. et al., 2016 (36)	Ethiopia	C/S	Community	548	≥60	261	287	MNA	9.3	28.3	62.4		Not reported
28.	Abate et al., 2020 (38)	Ethiopia	C/S	Community	662	≥65	240	422	MNA	26.6	73.4			One
29.	Abdu AO et al., 2020 (39)	Ethiopia	C/S	Community	592	≥65	291	301	MNA	15.7	51.7	32.6		One
30.	Abd Allah et al., 2020 (43)	Egypt	C/S	Hospital	194	≥60	107	87	MNA	33.0	51.5	15.5		Multi
31.	Rasha El-Desouky 2017 (44)	Egypt	C/S	Community	320	≥60	167	153	MNA	35.0	38.4	26.6		Multi
32.	Khater MS et al., 2011 (45)	Egypt	C/S	LTC	120	≥60	56	64	MNA	10.8	40.8	48.4		Multi
33.	F. Mahjoub et al., 2019 (68)	Tunis	C/S	Hospital	40	≥65	21	19	MNA	40.0	52.5	7.5		One
34.	Adebusoye, LA. et al., 2018 (52)	Nigeria	C/S	Hospital	624	≥60	246	378	MNA	2.24	32.69	65.06		One
35.	Abd-El-Gawad et al., 2014 (51)	Nigeria	Cohort	Hospital	131	≥60	67	64	MNA	40.46	52.67	6.87		Multi
36.	Mahfouz, E.M. et al., 2013 (46)	Egypt	C/S	Community	350	≥60	136	214	MNA	8.6	29.7	61.7		Multi
37.	Andia A, et al., 2019 (70)	Niger	C/S	Community	384	≥60	132	252	MNA	7.8	58.5	33.7		Multi

Abbreviations: BMI = Body mass index
CAR = Central Africa Republic
ROC= Republic of Congo
C/S: Cross-section
MNA: Mini Nutritional Assessment Tool
MUAC: mid-upper arm circumference
LTC: Long term care
MNA-SF: Mini Nutritional Assessment Tool- short Form
NCL: Nutritional Screening Checklist

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Supplementary Table 3. Summary of the quality of included studies

S/no	Author & year of publication	Country	External validity quality Score				Internal validity quality Score						Total Score	Study Quality
			1	2	3	4	5	6	7	8	9	10		
1.	Aganiba, B. et al., 2015 (62)	Ghana	0	1	1	1	1	1	1	1	1	1	9	High
2.	Apprey C. et al., 2019 (63)	Ghana	0	1	1	1	1	1	1	1	1	1	9	High
3.	Diendéré, J. et al, 2018 (35)	Burkina Faso	0	1	1	1	1	1	1	1	1	1	9	High
4.	Mkhize X. et al., 2013 (53)	South Africa	0	1	1	1	1	1	1	1	1	1	9	High
5.	MJ, C. et al, 2012 (69)	Lake Victoria Basin of East Africa	0	1	1	1	1	1	1	1	1	1	9	High
6.	Ijarotimi OS & Keshinro OO, 2004 (65)	Tanzania	0	0	0	1	1	1	1	1	1	1	7	Moderate
7.	Nyaruhucha CNM. et al, 2001 (66)	Tanzania	0	1	1	1	1	1	1	1	1	1	9	High
8.	Olayiwola O.et al, 2006 (47)	Nigria	0	1	1	1	1	1	1	1	1	1	9	High
9.	Rouvray C. et al., 2014 (61)	Central Africa (COR & ROC)	0	1	1	1	1	1	1	1	1	1	9	High
10.	Legesse M. et al., 2019 (34)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
11.	Wondiye K et al., 2019 (35)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
12.	Agbozo, F. et al., 2018 (64)	Ghana	0	1	1	1	1	1	1	1	1	1	9	High
13.	Alao M. et al., 2015 (48)	Nigeria	0	0	0	1	1	1	1	1	1	1	7	Moderate
14.	Tessfamichael, D. et al., 2014 (40)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
15.	Otitoola Oc. Et al., 2015 (57)	South Africa	0	0	0	0	1	1	1	1	1	1	6	Moderate
16.	Pilleron, S. et al., 2015 (58)	Central Africa (COR & ROC)	1	1	1	1	1	1	1	1	1	1	10	High

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17.	Esmayel EM. et al, 2013 (42)	Egypt	0	1	1	1	1	1	1	1	1	1	9	High
18.	Andre MB. et al., 2013 (59)	Democratic Republic of Congo	0	1	1	1	1	1	1	1	1	1	9	High
19.	Adebusoye L.A. et al., 2012 (49)	Nigeria	0	1	1	1	1	1	1	1	1	1	9	High
20.	Robb L. et al., 2017 (54)	South Africa	0	1	1	1	1	1	1	1	1	1	9	High
21.	El-Sherbiny N.A. et al., al.2016 (41)	Egypt	0	1	1	1	1	1	1	1	1	1	9	High
22.	Naidoo I. et al., 2015 (56)	South Africa	0	1	1	1	1	1	1	1	1	1	9	High
23.	Adhana, Z K et al., 2015 (37)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
24.	Adebusoye L. et al., 2019 (50)		0	0	1	1	1	1	1	1	1	1	8	Moderate
25.	Marias ML. et al., 2007 (55)	South Africa	0	1	1	1	1	1	1	1	1	1	9	High
26.	Hailemariam H. et al., 2016 (36)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
27.	Abate et al., 2020 (38)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
28.	Abdu AO et al., 2020 (39)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
29.	Abd Allah et al., 2020 (43)	Egypt	0	0	0	0	1	1	1	1	1	1	6	Moderate
30.	Rasha El-Desouky_2017 (44)	Egypt	0	0	1	1	1	1	1	1	1	1	8	Moderate
31.	Khater MS et al., 2011 (45)	Egypt	0	0	0	0	1	1	1	1	1	1	6	Moderate
32.	F. Mahjoub et al., 2019 (68)	Tunis	0	0	0	1	1	1	1	1	1	1	7	Moderate
33.	Adebusoye, LA. et al., 2018 (52)	Nigeria	0	0	0	0	1	1	1	1	1	1	6	Moderate
34.	Abd-El-Gawad et al., 2014 (51)	Nigeria	0	0	0	1	1	1	1	1	1	1	7	Moderate

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35.	Mahfouz, E.M. et al., 2013 (46)	Egypt	0	0	0	1	1	1	1	1	1	1	7	Moderate
36.	Andia A, et al., 2019 (70)	Niger	0	0	0	0	1	1	1	1	1	1	6	Moderate

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	2
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	3
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	3-4
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	3
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	3
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	3
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	4
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	3-5
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	4
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	4
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	4
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	3
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	4
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	5
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	5
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	5
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	5
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	4
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	N/A

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	5
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	6
Study characteristics	17	Cite each included study and present its characteristics.	6
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	7
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	7-8
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	8
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	8-9
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	9
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	9
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	N/A
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	N/A
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	10
	23b	Discuss any limitations of the evidence included in the review.	12
	23c	Discuss any limitations of the review processes used.	12
	23d	Discuss implications of the results for practice, policy, and future research.	12
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	3
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	3
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	13
Competing interests	26	Declare any competing interests of review authors.	13
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	13

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi:



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